

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization With Power BI & Tableau
Practical No 9

Aim :- Cleaning and Structuring Data for Analysis

A: Identify and Handle Missing or Inconsistent Data

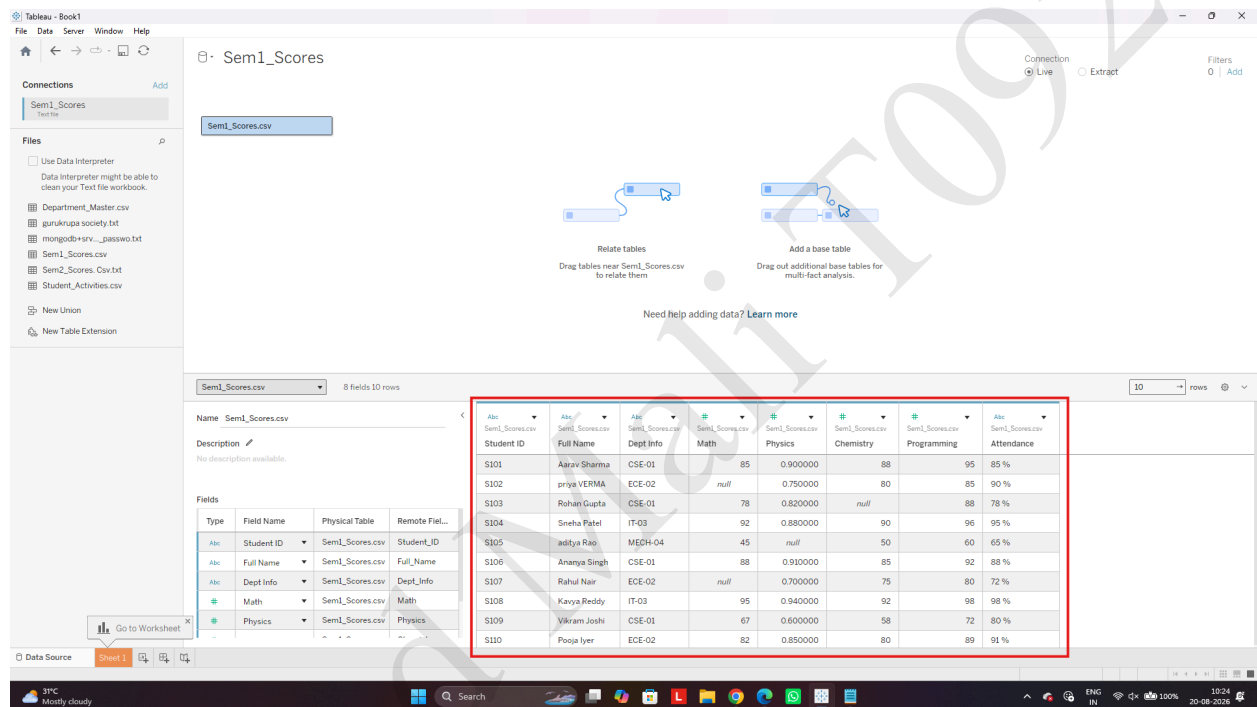
1. Open Tableau Desktop.

2. Under the Connect pane on the left, click Text file.

3. Select and open Sem1_Scores.csv.

4. In the Data Source preview table:

o Observe missing entries (Null) in Math, Physics, and Chemistry.



Step 2: Clean Inconsistent String Fields (Convert Text Percentages to Numeric Measures)

1. Navigate to Sheet 1 (bottom-left tab).

2. Right-click an empty space in the Data Pane (left sidebar) --> select Create Calculated Field....

3. Create the cleaned measure for Physics:

o Field Name: Physics_Clean o Tableau Formula: Code snippet

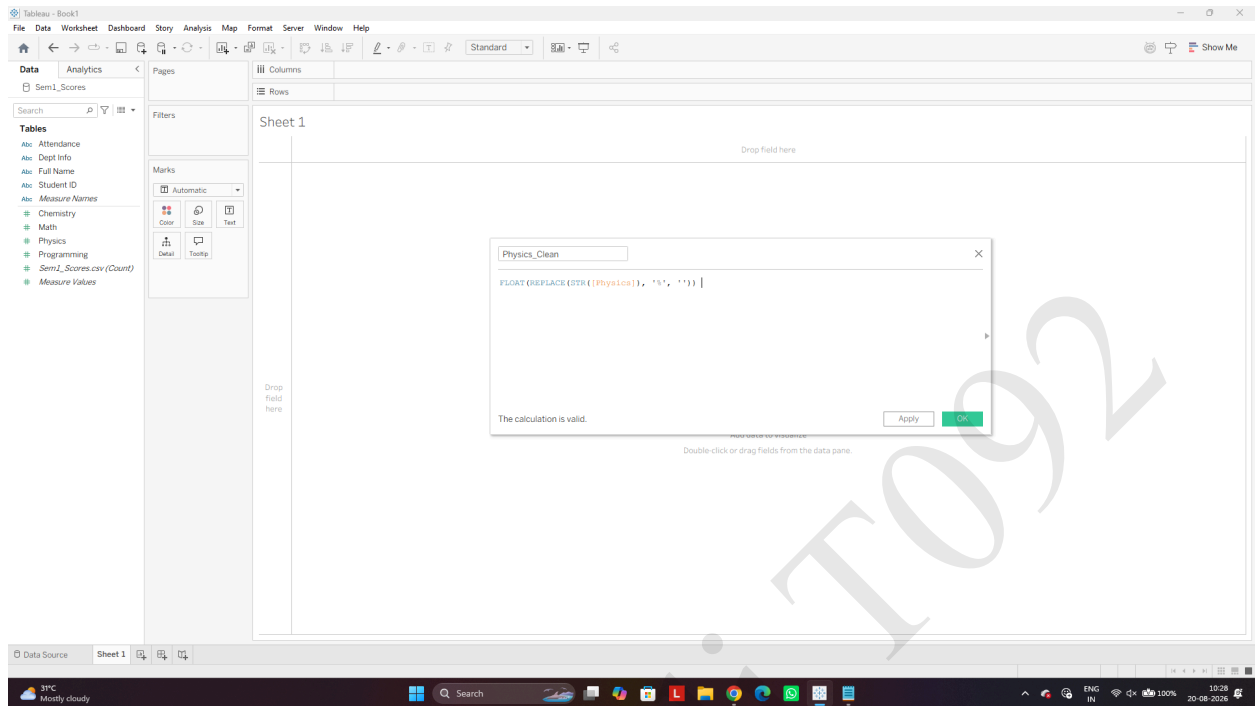
FLOAT(REPLACE(STR([Physics]), '%', ''))

o Click OK.

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4. Create the cleaned measure for Attendance:

o Right-click the Data Pane --> Create Calculated Field....

o Field Name: Attendance_Clean

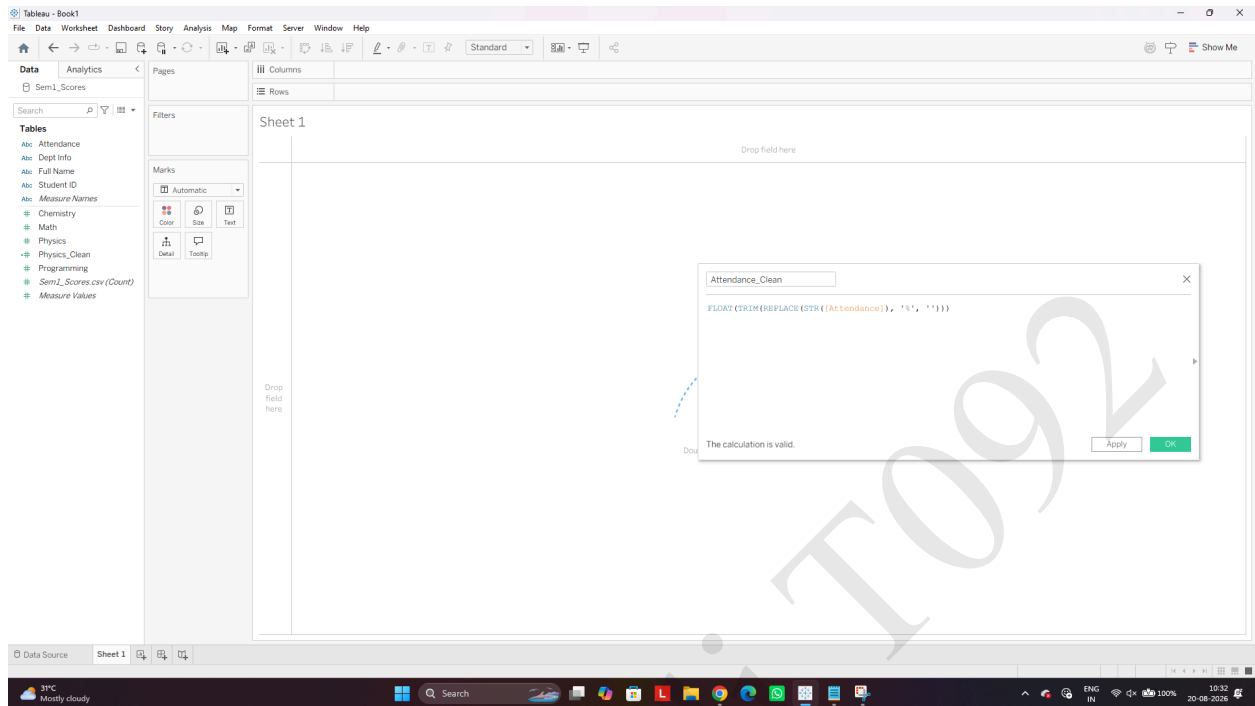
o Tableau Formula: Code snippet `FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))`

o Click OK.

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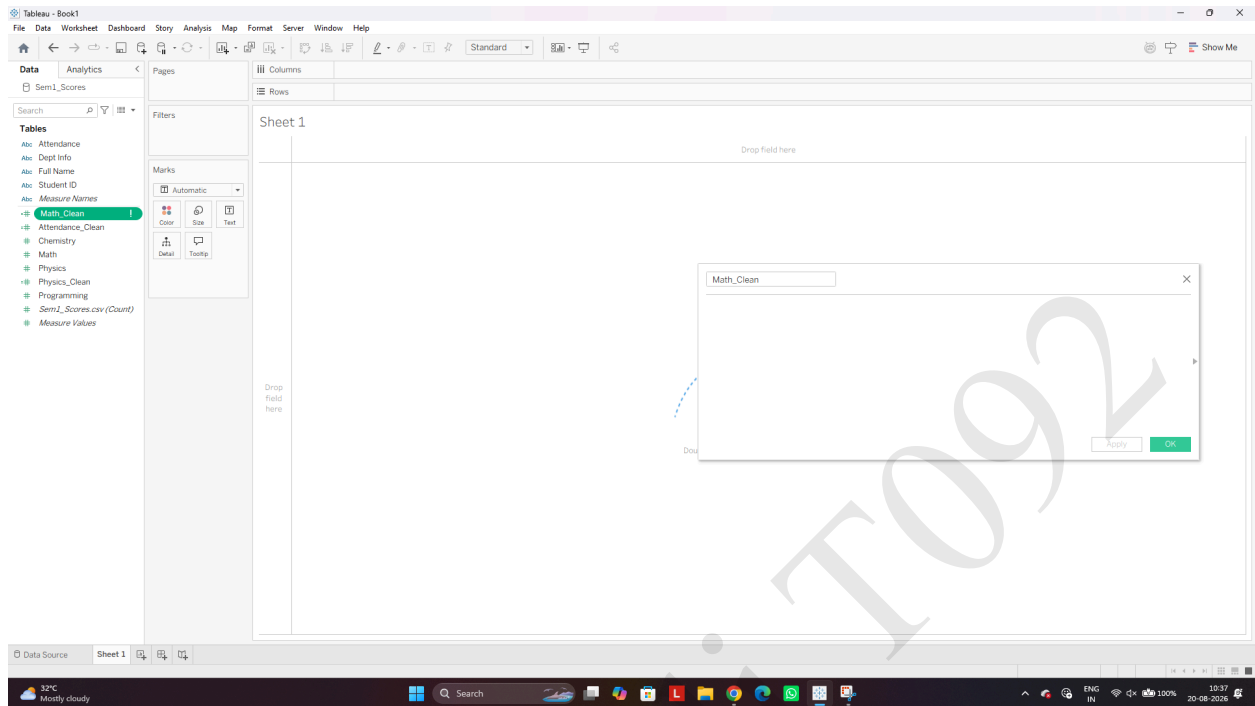
Step 3: Handle Missing (Null) Numeric Values via Imputation

1. Handle missing entries in Math:

- o Right-click the Data Pane --> Create Calculated Field....
- o Field Name: Math_Clean
- o Tableau Formula: Code snippet IFNULL([Math], 0)
- o Click OK.

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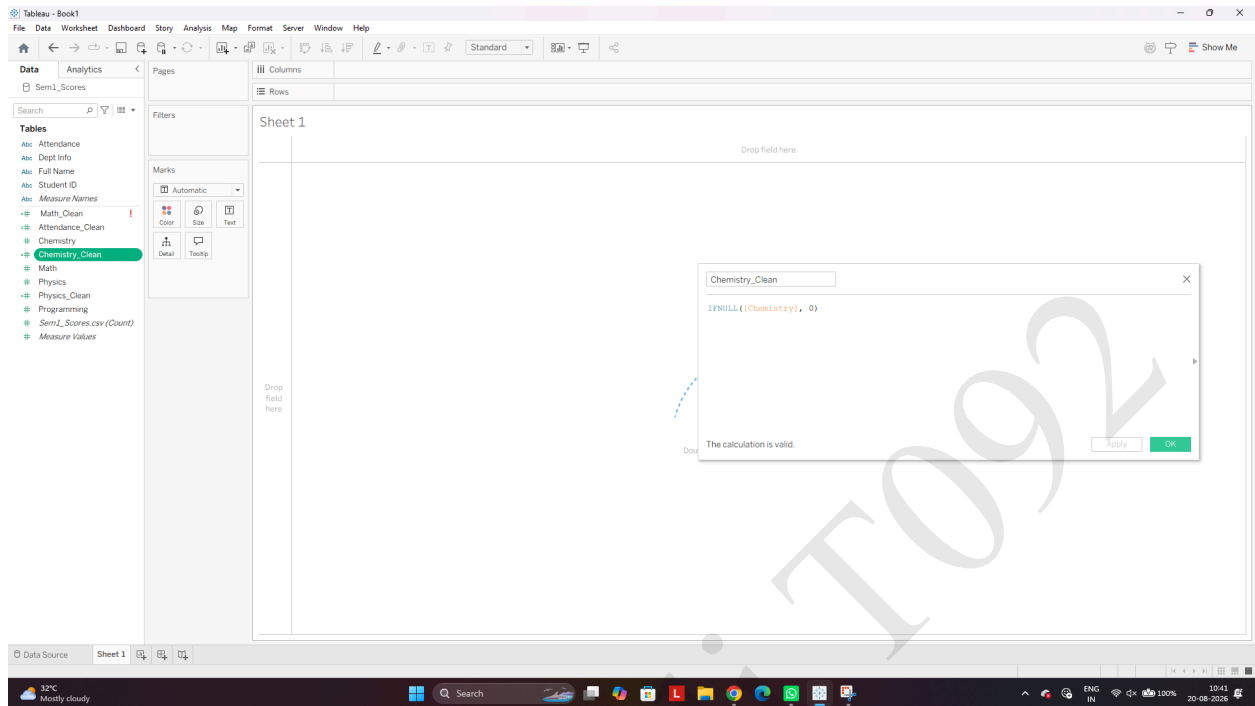


2. Handle missing entries in Chemistry:

- o Right-click the Data Pane --> Create Calculated Field....
- o Field Name: Chemistry_Clean
- o Tableau Formula: Code snippet IFNULL([Chemistry], 0)
- o Click OK.

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Step 4: Impute Missing Values in Physics (Optional Handling for Student S105)

1. Student S105 contains a Null value in Physics. Combine string stripping with null handling:

o Edit Physics_Clean (right-click Physics_Clean --> Edit...):

o Updated Tableau Formula: Code snippet

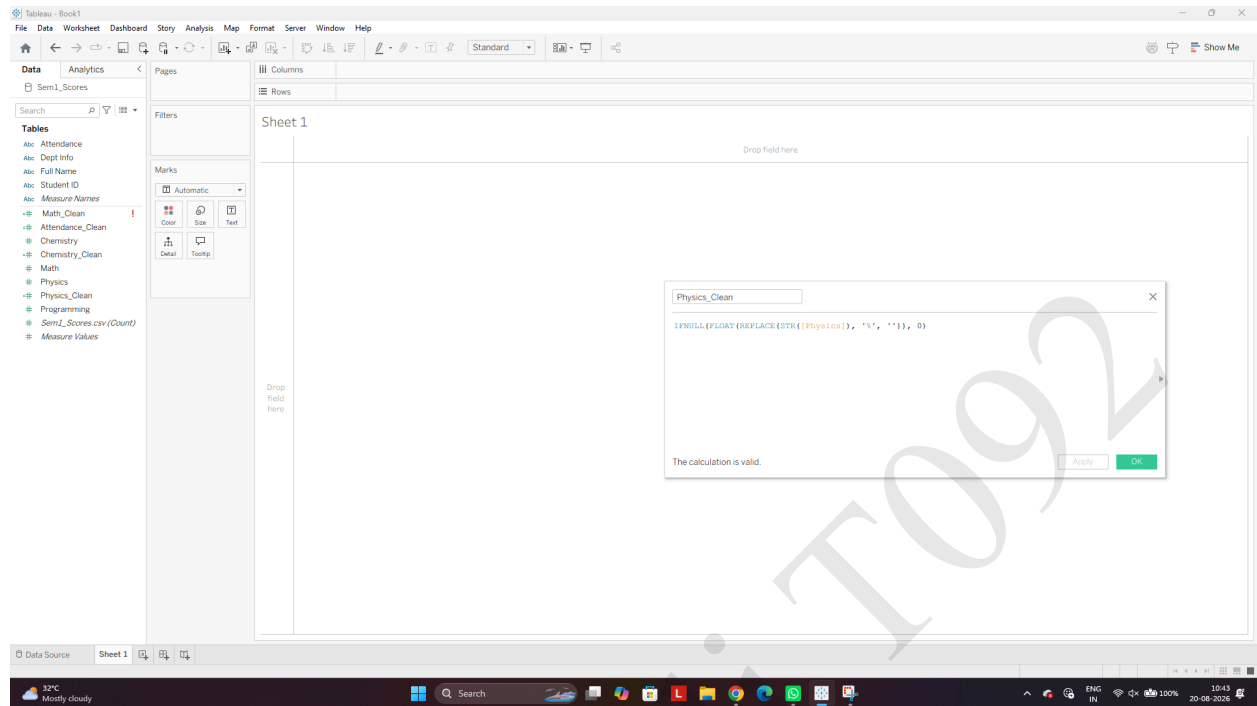
IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)

o Click OK

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B: Convert Wide Data into Tall Format

Step 1: Open the Data Source Grid

- 1. In Tableau, navigate to the Data Source tab at the bottom-left corner.**
- 2. Locate the four horizontal subject score columns:**
 - o Math**
 - o Physics**
 - o Chemistry**
 - o Programming**

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The screenshot shows the Tableau Desktop interface with the 'Sem1_Scores' data source loaded. The 'Fields' shelf contains the following fields: Student ID, Full Name, Dept Info, Math, Physics, Chemistry, Programming, Attendance, Physics_Clean, and Attendance_Clean. The 'Columns' shelf is empty. The 'Rows' shelf contains 'Student ID'. The 'Marks' shelf contains 'Full Name'. The 'Context' shelf is empty. The 'Filters' shelf is empty.

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance	Physics_Clean	Attendance_Clean
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %	0.900000	85.0000
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %	0.750000	90.0000
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %	0.820000	78.0000
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %	0.880000	95.0000
S105	aditya Rao	MECH-04	45	null	50	60	65 %	0.000000	65.0000
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %	0.910000	88.0000
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %	0.700000	72.0000
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98 %	0.940000	98.0000
S109	Vikram Joshi	CSE-01	67	0.600000	58	72	80 %	0.600000	80.0000
S110	Pooja Iyer	ECE-02	82	0.850000	80	89	91 %	0.850000	91.0000

Step 2: Multi-Select and Pivot Columns

1. Click the header of the Math column.
2. Hold down Ctrl (or Cmd on Mac) and click Physics, Chemistry, and Programming so all four columns are highlighted in blue.

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Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv

gunakrupa society.txt

mongodb+srv_..._passwo.txt

Sem1_Scores.csv

Sem2_Scores.Csv.txt

Student_Activities.csv

New Union

New Table Extension

Sem1_Scores

Relate tables

Drag tables near Sem1_Scores.csv to relate them

Add a base table

Drag out additional base tables for multi-fact analysis.

Need help adding data? [Learn more](#)

Sem1_Scores.csv 12 fields 10 rows

Name Sem1_Scores.csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
+	Math	Sem1_Scores.csv	Math
+	Physics	Sem1_Scores.csv	Physics
+	Chemistry	Sem1_Scores.csv	Chemistry
+	Programming	Sem1_Scores.csv	Programming
+	Attendance	Sem1_Scores.csv	Attendance

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance	Physics_Clean	Attendance_Clean
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %	0.900000	85.0000
S102	priya VERMA	ECE-02	78	0.750000	80	85	90 %	0.750000	90.0000
S103	Rohan Gupta	CSE-01	78	0.820000	88	96	78 %	0.820000	78.0000
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %	0.880000	95.0000
S105	aditya Rao	MECH-04	45	0.600000	50	60	65 %	0.600000	65.0000
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %	0.910000	88.0000
S107	Rahul Nair	ECE-02	75	0.700000	75	80	72 %	0.700000	72.0000
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98 %	0.940000	98.0000
S109	Vikram Joshi	CSE-01	67	0.600000	58	72	80 %	0.600000	80.0000
S110	Pooja Iyer	ECE-02	82	0.850000	80	89	91 %	0.850000	91.0000

3. Right-click on any of the highlighted column headers.

4. Select Pivot

Tableau - Book1

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gunakrupa society.txt

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Sem1_Scores.csv

Sem2_Scores.Csv.txt

Student_Activities.csv

New Union

New Table Extension

Sem1_Scores

Relate tables

Drag tables near Sem1_Scores.csv to relate them

Add a base table

Drag out additional base tables for multi-fact analysis.

Need help adding data? [Learn more](#)

Sem1_Scores.csv 12 fields 10 rows

Name Sem1_Scores.csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
+	Math	Sem1_Scores.csv	Math
+	Physics	Sem1_Scores.csv	Physics
+	Chemistry	Sem1_Scores.csv	Chemistry
+	Programming	Sem1_Scores.csv	Programming
+	Attendance	Sem1_Scores.csv	Attendance

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance	Physics_Clean	Attendance_Clean
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %	0.900000	85.0000
S102	priya VERMA	ECE-02	78	0.750000	80	85	90 %	0.750000	90.0000
S103	Rohan Gupta	CSE-01	78	0.820000	88	96	78 %	0.820000	78.0000
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %	0.880000	95.0000
S105	aditya Rao	MECH-04	45	0.600000	50	60	65 %	0.600000	65.0000
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %	0.910000	88.0000
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S108	Kavya Reddy	IT-03	95	0.940000	92	98	98 %	0.940000	98.0000
S109	Vikram Joshi	CSE-01	67	0.600000	58	72	80 %	0.600000	80.0000
S110	Pooja Iyer	ECE-02	82	0.850000	80	89	91 %	0.850000	91.0000

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step 3: Rename the Pivoted Fields Tableau replaces the four columns with two consolidated columns at the end of the table:

1. Double-click the header named Pivot Field Names and type: Subject

2. Double-click the header named Pivot Field Values and type: Score

Tableau - Book1

File Data Server Window Help

Connections Add

Sem1_Scores Text file

Files

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Department_Master.csv
gurukrupa society.txt
mongodb+srv..._passwo.txt
Sem1_Scores.csv
Sem2_Scores.csv.txt
Student_Activities.csv

New Union
New Table Extension

Sem1_Scores

Sem1_Scores.csv

Connection Live Extract Filters 0 Add

Relate tables
Drag tables near Sem1_Scores.csv to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? Learn more

Sem1_Scores.csv 10 fields 40 rows

Name Sem1_Scores.csv
Description No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
Math	Math	Sem1_Scores.csv	Math
Subject	Pivot	Pivot	Pivot Field Na...
Score	Pivot Field Val...	Calculation	Calculation_2...
Attendance	Calculation	Calculation_2...	

Student ID	Full Name	Dept Info	Math	Pivot	Pivot Field Names	Pivot Field Values	Physics_Clean	Attendance_Clean	Math_Clean
S101	Aarav Sharma	CSE-01	85	Attendance	85 %	null	null	null	null
S101	Aarav Sharma	CSE-01	85	Chemistry	88	null	null	null	null
S101	Aarav Sharma	CSE-01	85	Physics	90%	null	null	null	null
S101	Aarav Sharma	CSE-01	85	Programming	95	null	null	null	null
S102	priya VERMA	ECE-02	null	Attendance	90 %	null	null	null	null
S102	priya VERMA	ECE-02	null	Chemistry	80	null	null	null	null
S102	priya VERMA	ECE-02	null	Physics	75%	null	null	null	null
S102	priya VERMA	ECE-02	null	Programming	85	null	null	null	null
S103	Rohan Gupta	CSE-01	78	Attendance	78 %	null	null	null	null
S103	Rohan Gupta	CSE-01	78	Chemistry	null	null	null	null	null

Tableau - Book1

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Sem1_Scores.csv
Sem2_Scores.csv.txt
Student_Activities.csv

New Union
New Table Extension

Sem1_Scores

Sem1_Scores.csv

Connection Live Extract Filters 0 Add

Relate tables
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Sem1_Scores.csv 10 fields 40 rows

Name Sem1_Scores.csv
Description No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
Math	Math	Sem1_Scores.csv	Math
Subject	Pivot	Pivot	Pivot Field Na...
Score	Pivot Field Val...	Calculation	Calculation_2...
Attendance	Calculation	Calculation_2...	

Student ID	Full Name	Dept Info	Math	Subject	Pivot Field Values	Physics_Clean	Attendance_Clean	Math_Clean
S101	Aarav Sharma	CSE-01	85	Attendance	85 %	null	null	null
S101	Aarav Sharma	CSE-01	85	Chemistry	88	null	null	null
S101	Aarav Sharma	CSE-01	85	Physics	90%	null	null	null
S101	Aarav Sharma	CSE-01	85	Programming	95	null	null	null
S102	priya VERMA	ECE-02	null	Attendance	90 %	null	null	null
S102	priya VERMA	ECE-02	null	Chemistry	80	null	null	null
S102	priya VERMA	ECE-02	null	Physics	75%	null	null	null
S102	priya VERMA	ECE-02	null	Programming	85	null	null	null
S103	Rohan Gupta	CSE-01	78	Attendance	78 %	null	null	null
S103	Rohan Gupta	CSE-01	78	Chemistry	null	null	null	null

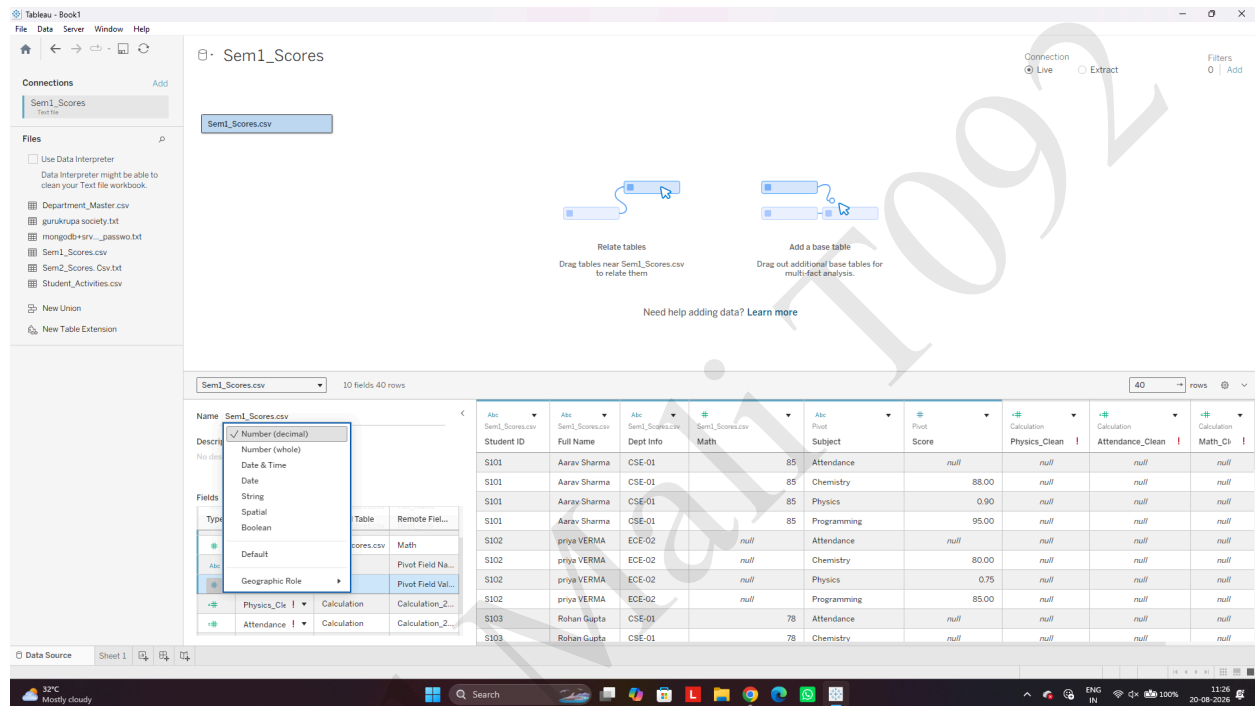
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Step 4: Clean and Format the Tall Score Field Because raw Physics contained % symbols, the new combined Score column defaults to text (Abc). Convert it to a numeric measure:

- 1. Click the Abc icon directly above the Score column header.**
- 2. Select Number (decimal) from the dropdown list.**



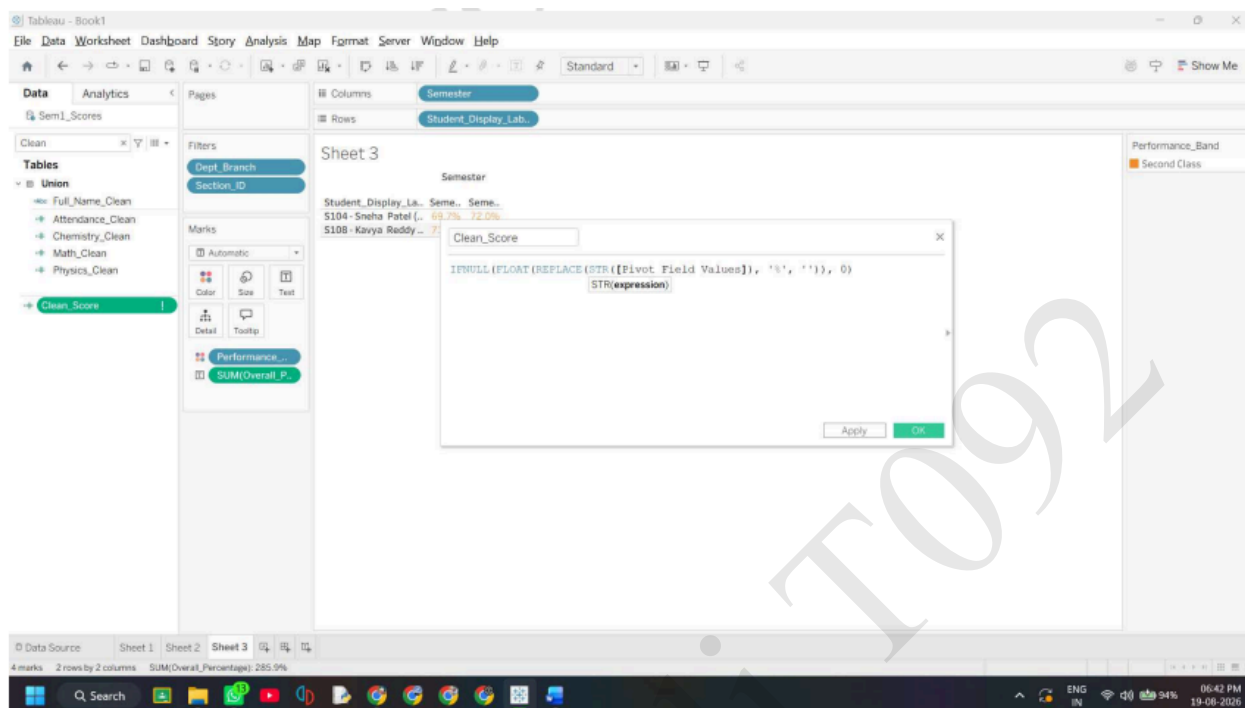
- 3. Alternatively, create a calculated field inside Sheet 1 named Clean_Score:**

Code snippet

IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)

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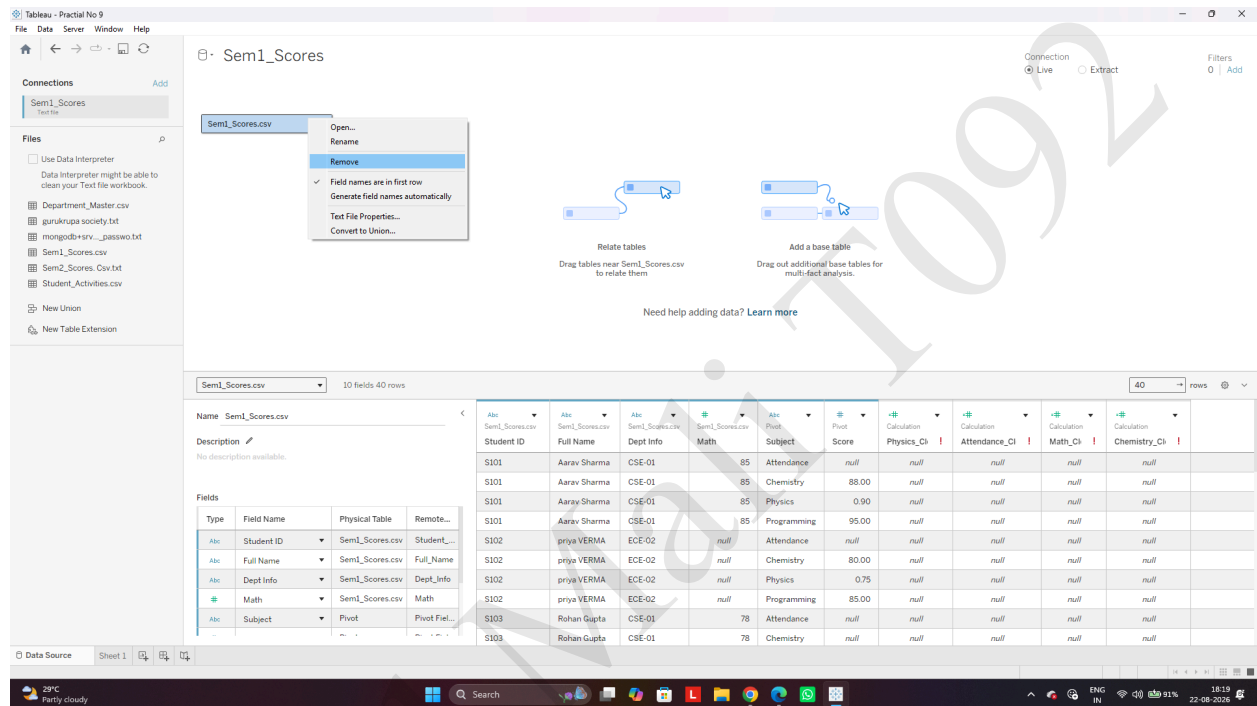
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C: Perform Union of Multiple Files

Step 1: Open the Data Source Canvas

1. Open Tableau Desktop and go to the Data Source tab in the bottom-left corner.
2. If any single table is already on the canvas, right-click it and select Remove to start with a clean canvas.

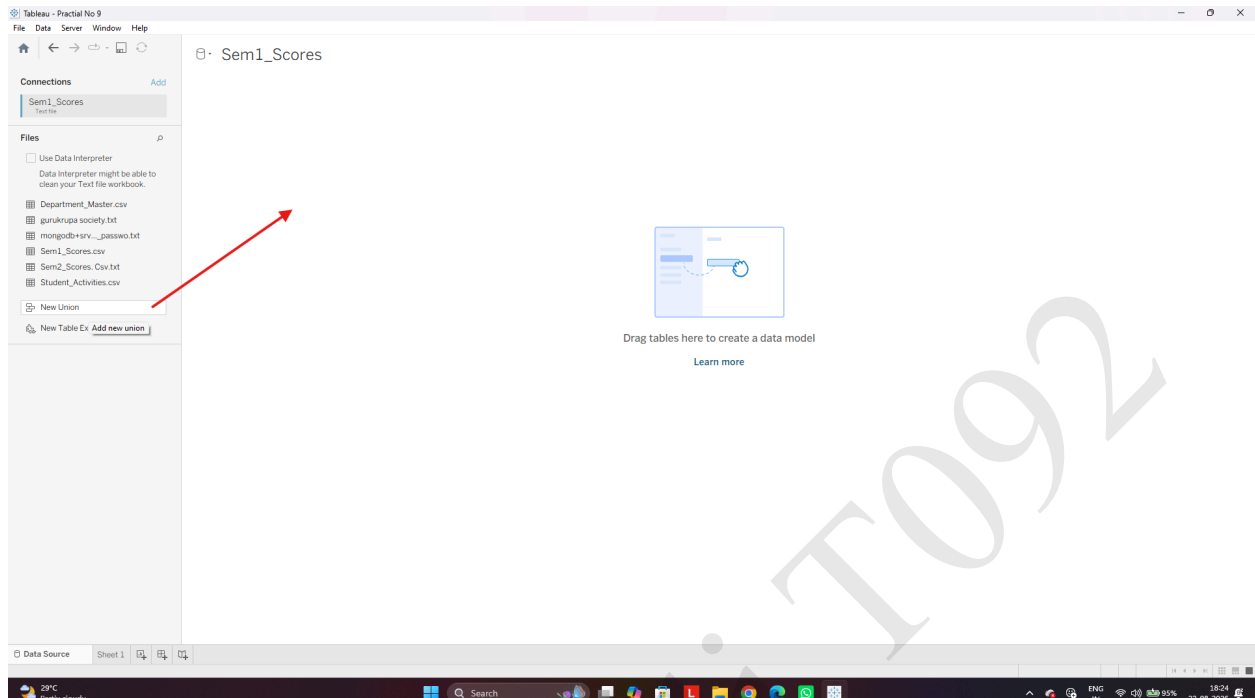


Step 2: Create a Multi-File Union

1. Look at the left sidebar under the Files / Connections pane.
2. Drag the New Union option onto the empty canvas area.

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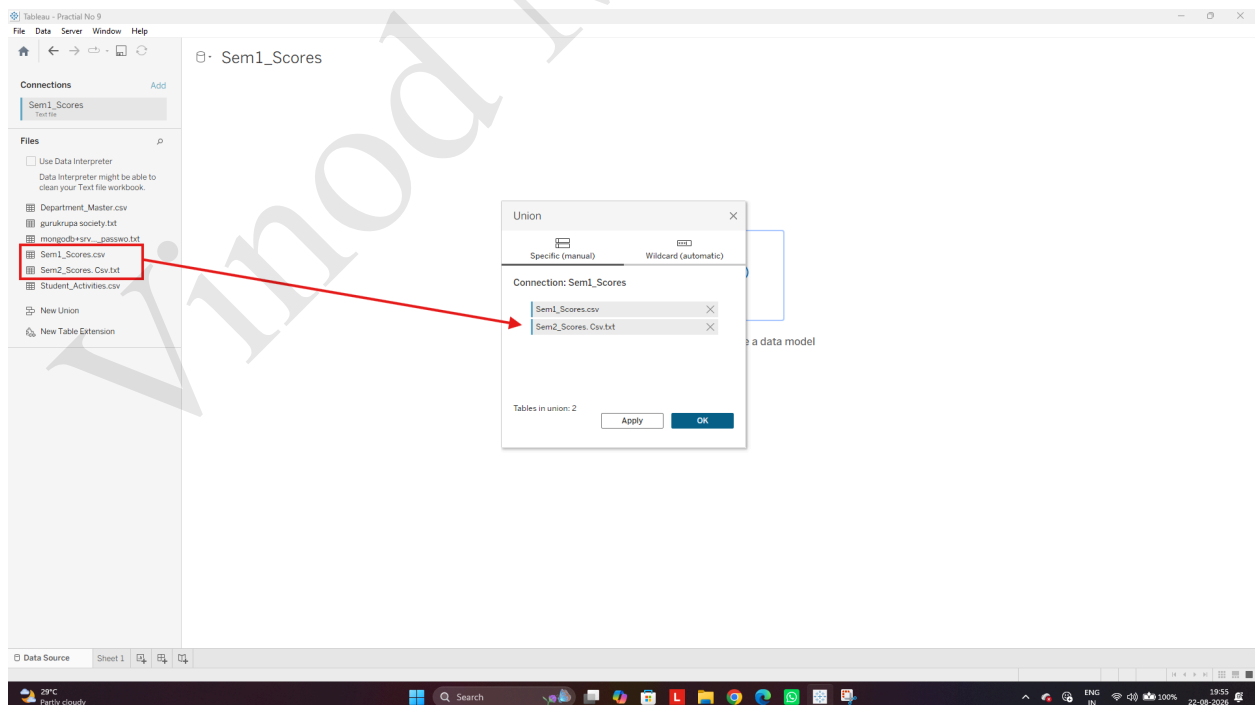


3. In the pop-up Union dialog box:

o Drag Sem1_Scores.csv from the left panel into the box.

o Drag Sem2_Scores.csv from the left panel into the same box.

o Verify that both filenames are listed in the Union box. 4. Click Apply, then click OK.



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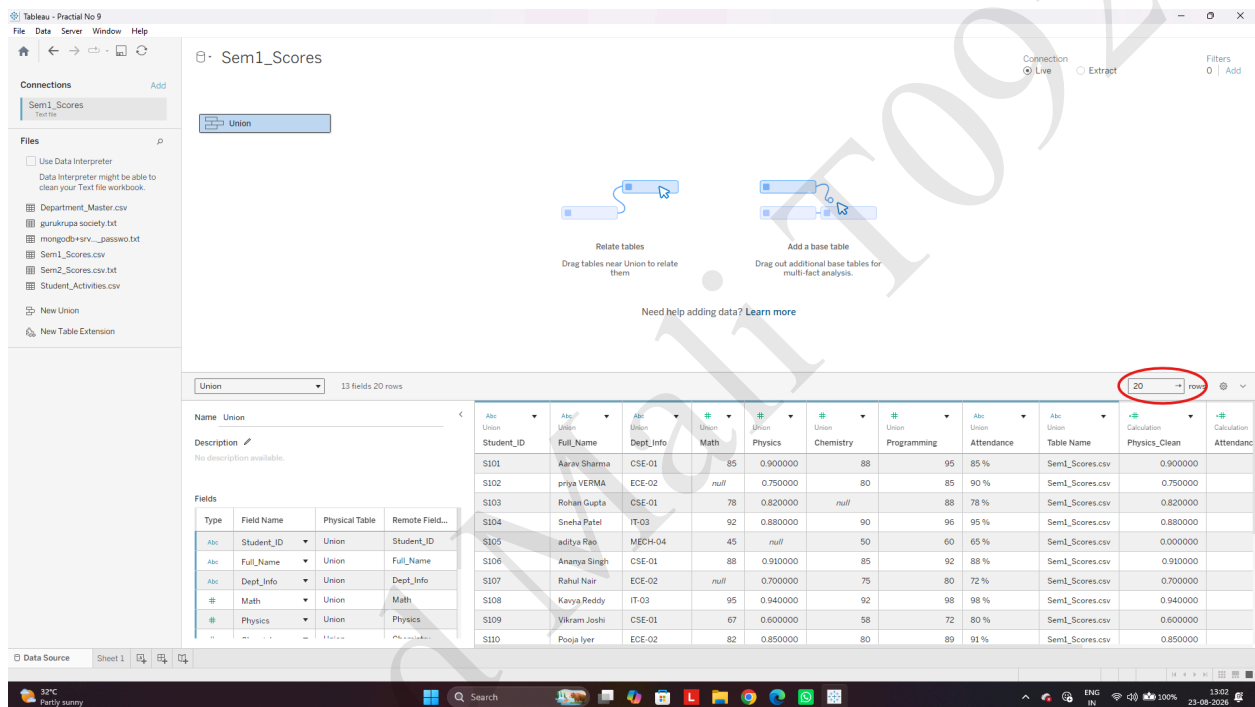
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5. The canvas table icon will now display with a plus symbol (+), representing the consolidated union.

Step 3: Verify the Merged Data Grid

1. Review the data grid at the bottom of the screen.

2. Confirm the row count: the total number of rows increases from 10 to 20 rows (10 records for Semester 1 and 10 records for Semester 2).



3. Scroll to the far right of the grid to verify the generated system column:

o Table Name: Displays "Sem1_Scores.csv" for the first 10 rows and "Sem2_Scores.csv" for the remaining 10 rows.

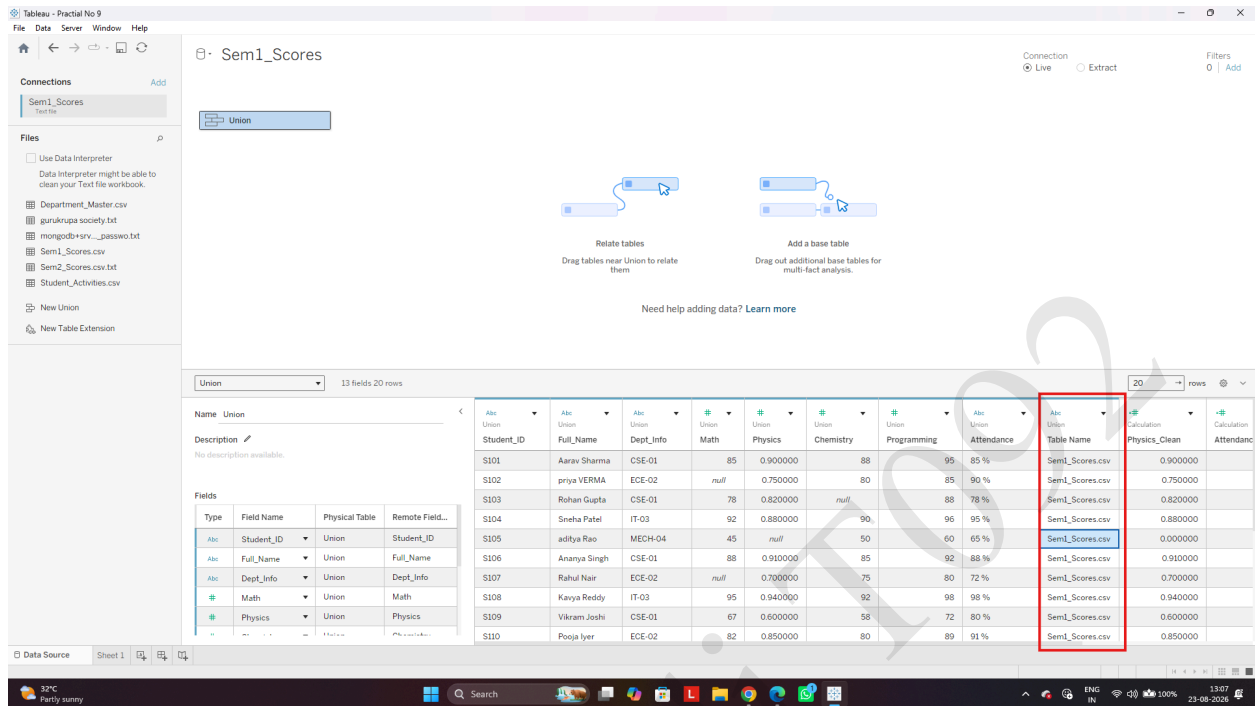
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Step 4: Create a Clean Semester Identifier Dimension

1. Click on Sheet 1 at the bottom-left.
2. In the Data Pane (left sidebar), right-click in an empty space select Create Calculated Field....
3. Set the Field Name to: Semester
4. Enter the following formula:

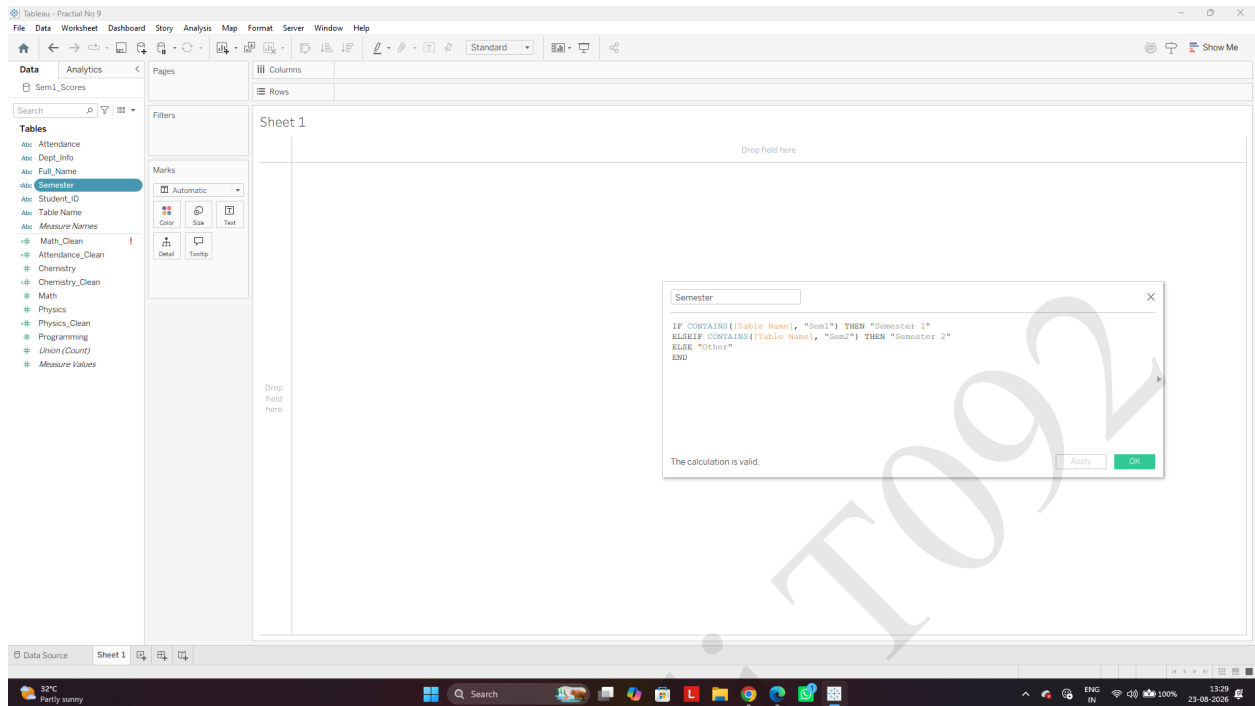
Code snippet

```
IF CONTAINS([Table Name], "Sem1") THEN "Semester 1" ELSEIF
CONTAINS([Table Name], "Sem2") THEN "Semester 2" ELSE "Other" END
```

5. Click OK.

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D: Execute Cross-Database Joins

Step 1: Stay on the Top Canvas

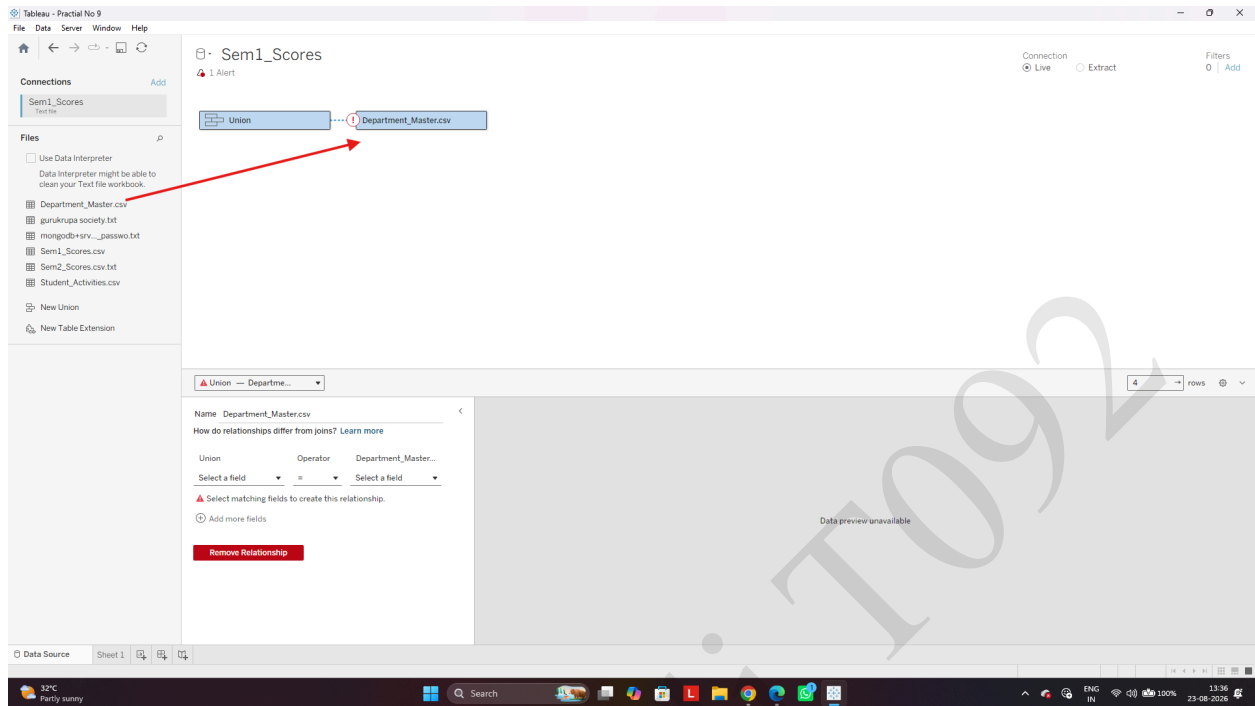
- Leave your Union table box as it is on the canvas.

Step 2: Add and Link Department_Master.csv

1. In the left sidebar under Files / Connections, find Department_Master.csv.
2. Drag Department_Master.csv onto the canvas and drop it to the right of your Union box.
3. A flexible line (relationship / join noodle) will connect the two boxes.

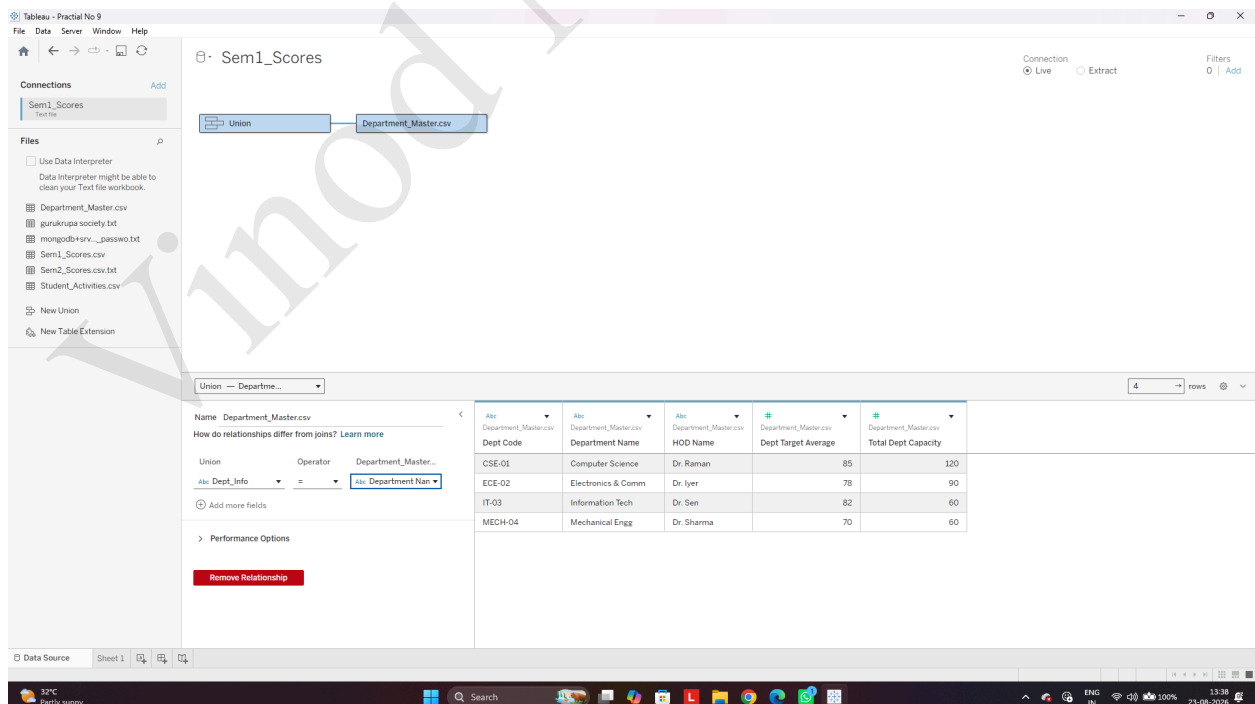
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4. If the relationship dialog opens (or click the relationship line):

- Left Table Field (Union): Select Dept_Info
- Right Table Field (Department_Master): Select Dept_Code
- Operator: =



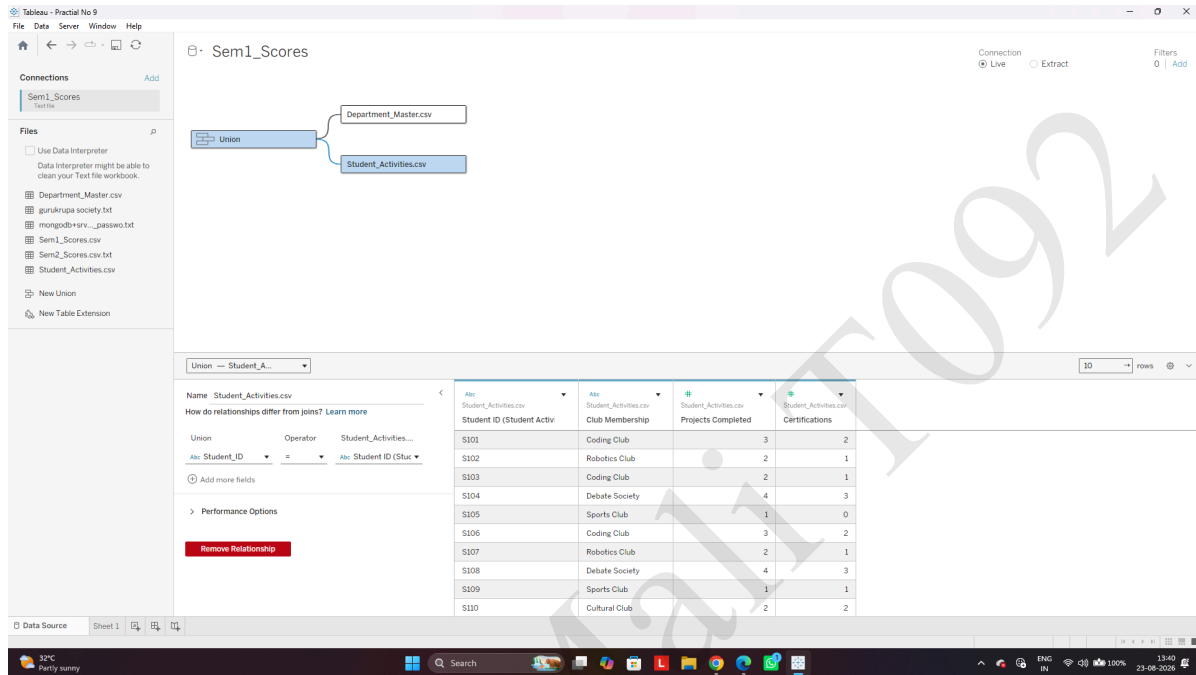
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Step 3: Add and Link Student_Activities.csv

1. From the left sidebar, drag Student_Activities.csv onto the canvas.
2. Drop it to the right of the Union box.



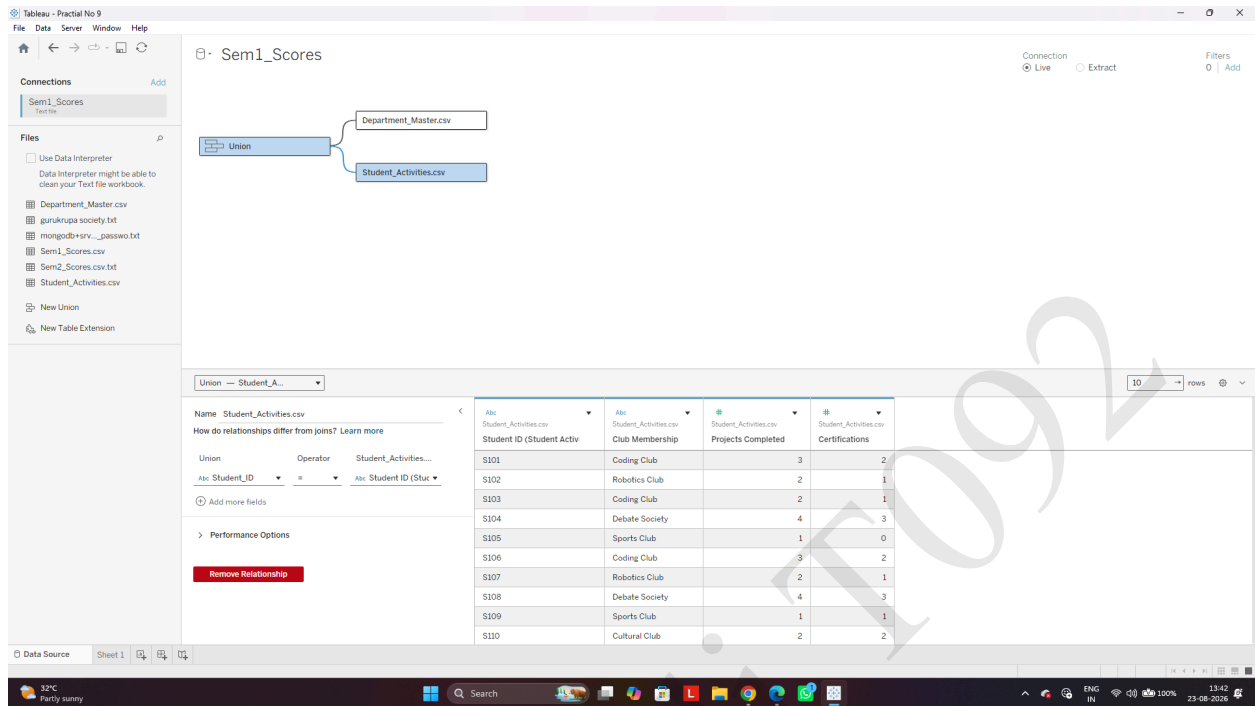
3. In the relationship dialog:

- Left Table Field (Union): Select Student_ID
- Right Table Field (Student_Activities): Select Student_ID
- Operator: =

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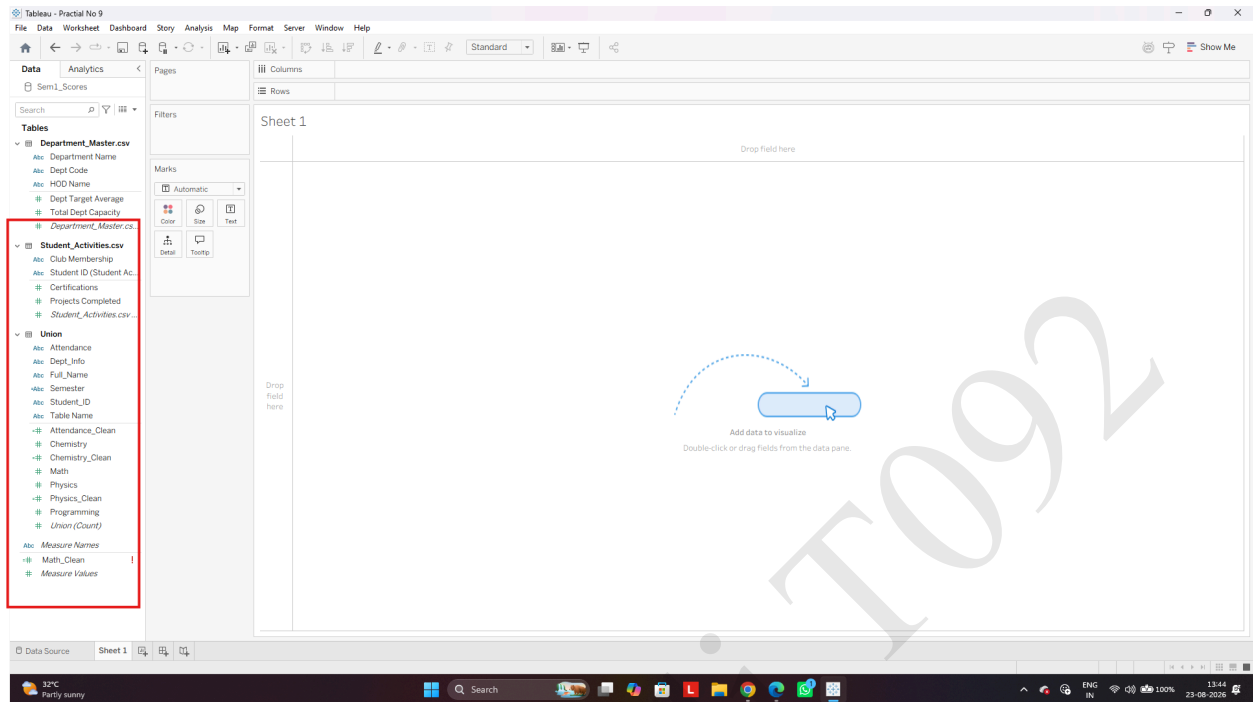


Step 4: Verify in Sheet 1

1. Click on Sheet 1 at the bottom-left.
2. In the Data Pane on the far left, you will now see all three tables organized with their respective fields:
 - Union: Student_ID, Full_Name, Dept_Info, Math, Physics, Chemistry, Programming, Attendance, Table Name
 - Department_Master: Dept_Code, Department_Name, HOD_Name, Dept_Target_Average, Total_Dept_Capacity
 - Student_Activities: Club_Membership, Projects_Completed, Certifications

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E: Restructure Data for Better Visualization

Step 1: Calculate Total Marks and Scaled Percentage

1. Navigate to Sheet 1 in Tableau

2. In the Data Pane on the left, right-click an empty space --> select Create Calculated Field....

3. Create the student's total marks field:

o Field Name: Total_Score

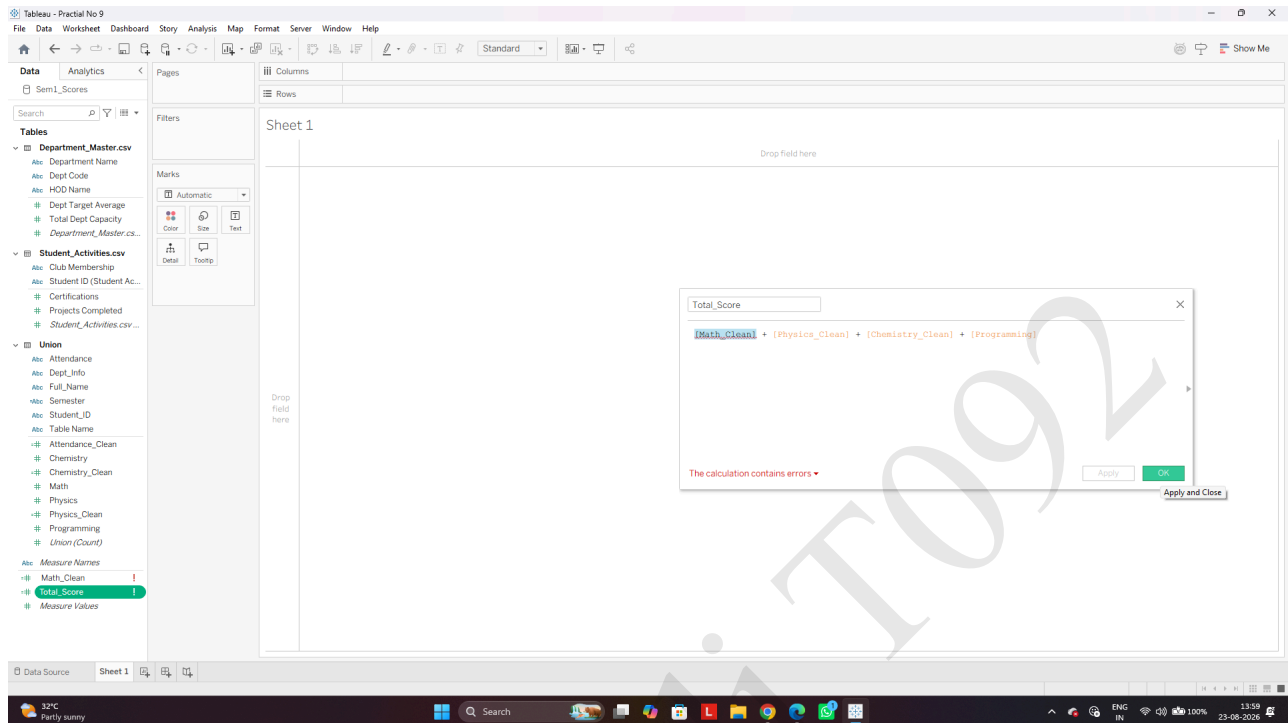
o Tableau Formula: Code snippet [Math_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Programming]

o Click OK.

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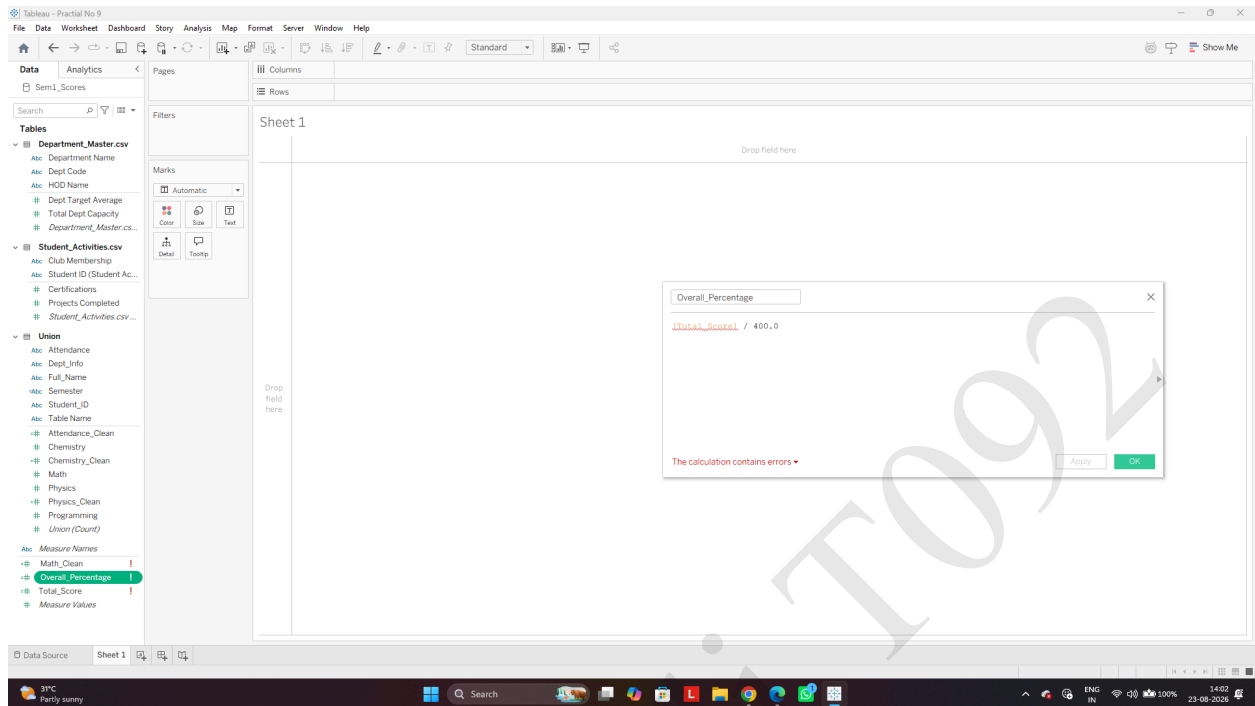


4. Create the scaled overall percentage field:

- o Right-click in the Data Pane --> select Create Calculated Field....
- o Field Name: Overall_Percentage
- o Tableau Formula: Code snippet $[Total_Score] / 400.0$
- o Click OK.

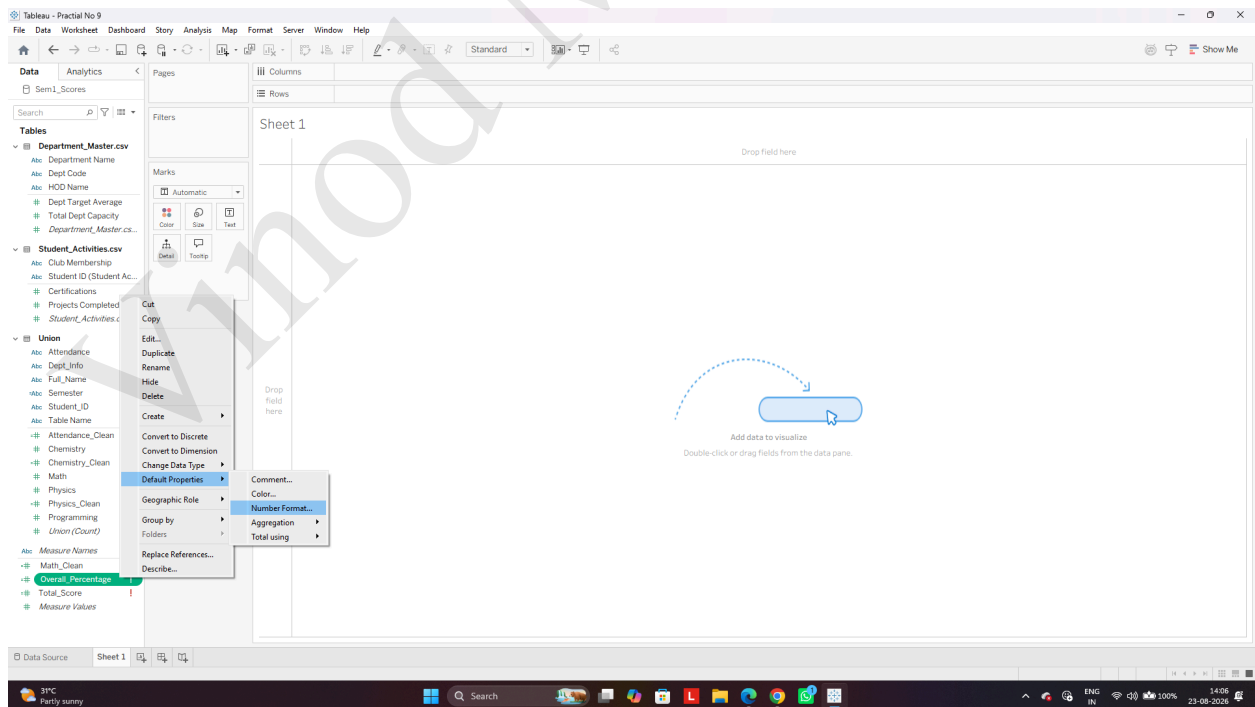
Name :- Vinod Manohar Mali
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5. Set the default percentage display formatting:

- o In the Data Pane, right-click Overall_Percentage -->Default Properties
- >Number Format....

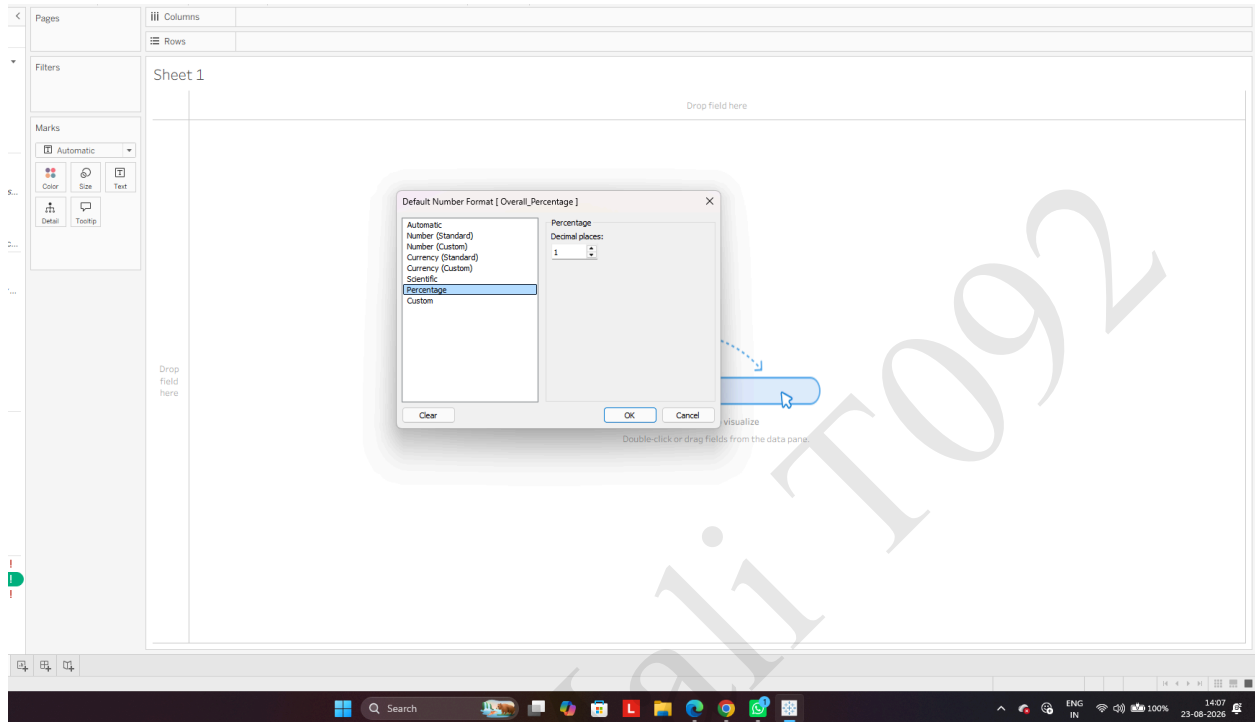


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- o Select Percentage and set decimal places to 1 (e.g., 85.5%).
- o Click OK



Step 2: Create Categorical Performance Bands (Grading Tiers)

1. Right-click in the Data Pane -->select Create Calculated Field....

2. Create the academic grading tier field:

- o Field Name: Performance_Band

- o Tableau Formula:

Code snippet

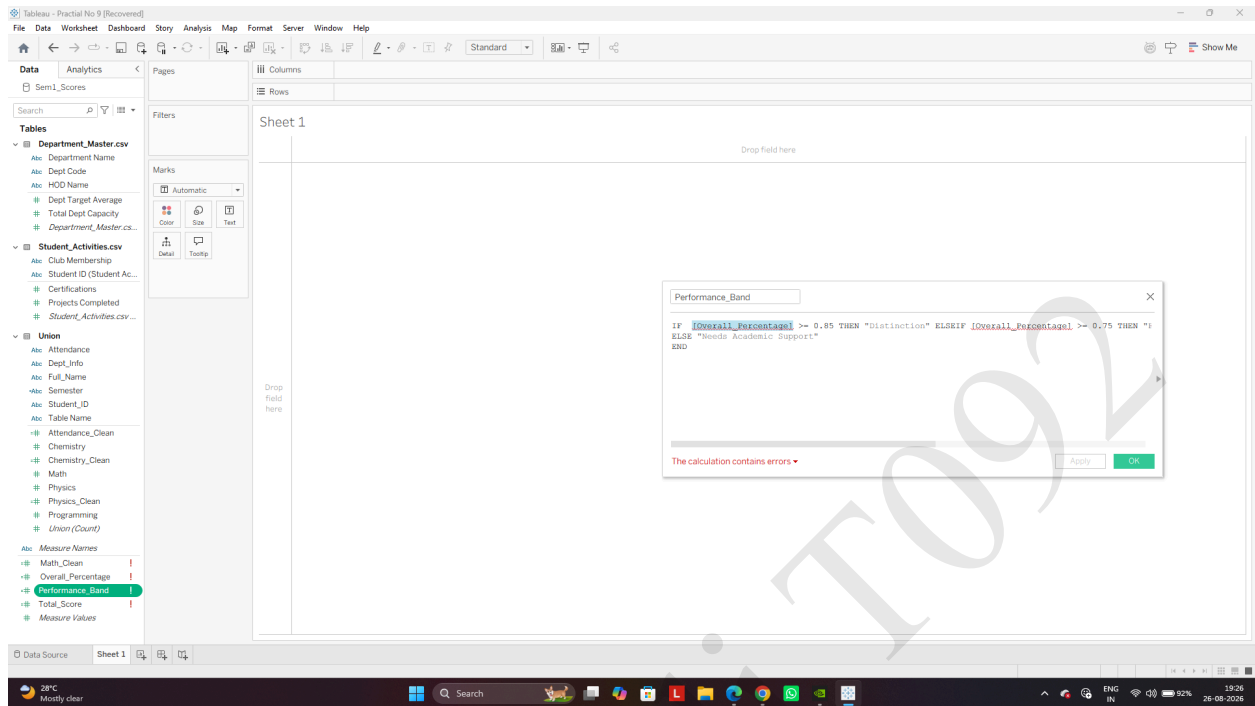
```
IF [Overall_Percentage] >= 0.85 THEN "Distinction" ELSEIF  
[Overall_Percentage] >= 0.75 THEN "First Class" ELSEIF  
[Overall_Percentage] >= 0.60 THEN "Second Class"  
ELSE "Needs Academic Support" END
```

- o Click OK.

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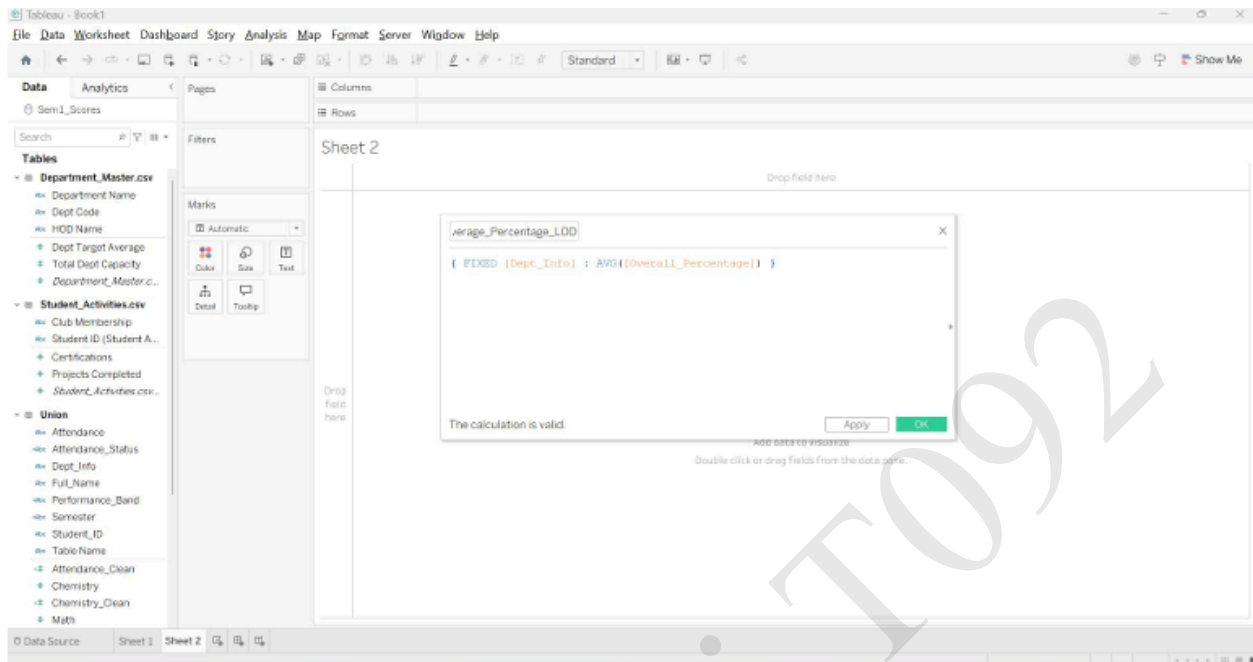
F: Handle Different Levels of Detail in Datasets

Step 1: Compute Department-Level Benchmark using {FIXED} LOD

- 1. In Tableau, navigate to Sheet 1 (or create a new worksheet named Sheet 2).**
- 2. In the Data Pane on the left, right-click an empty space --> select Create Calculated Field....**
- 3. Create the department-wide benchmark field:**
 - o Field Name: Dept_Average_Percentage_LOD**
 - o Tableau Formula: Code snippet { FIXED [Dept_Info] : AVG([Overall_Percentage]) }**
 - o Click OK.**

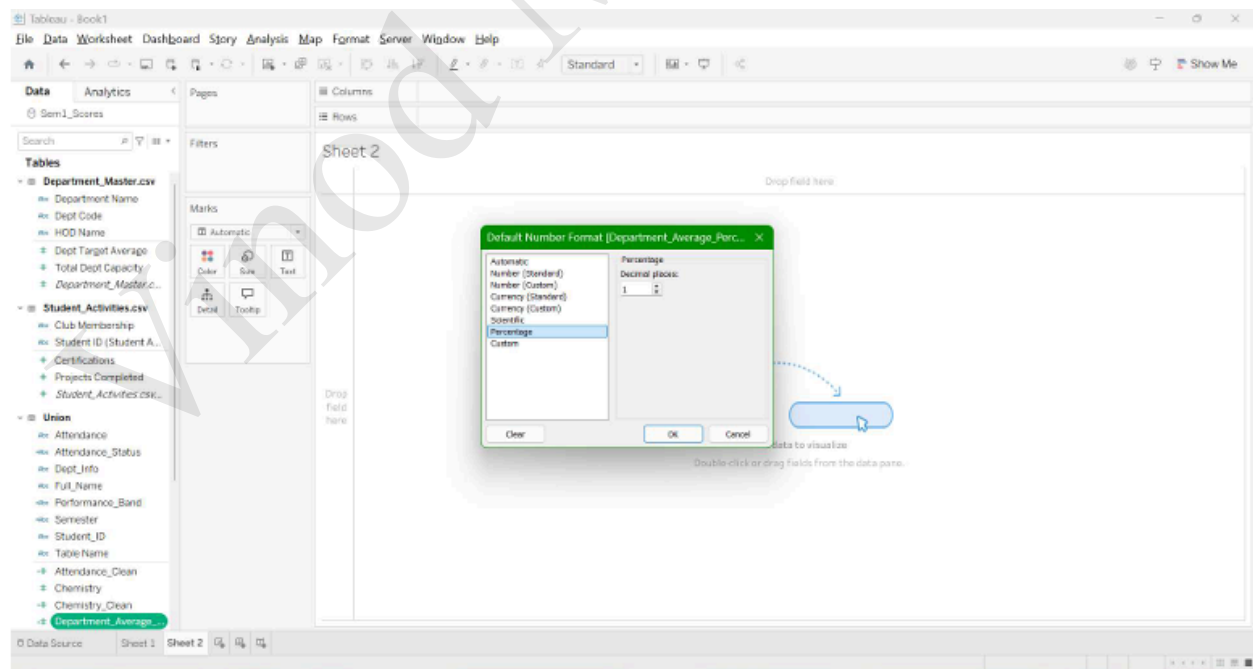
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4. Set default formatting:

o Right-click Dept_Average_Percentage_LOD --> Default Properties --> Number Format... --> select Percentage (1 decimal place) --> OK.



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Step 2: Compute Student Multi-Semester Average using {FIXED} LOD

1. Right-click in the Data Pane --> select Create Calculated Field....

2. Create the cumulative student benchmark across all semesters:

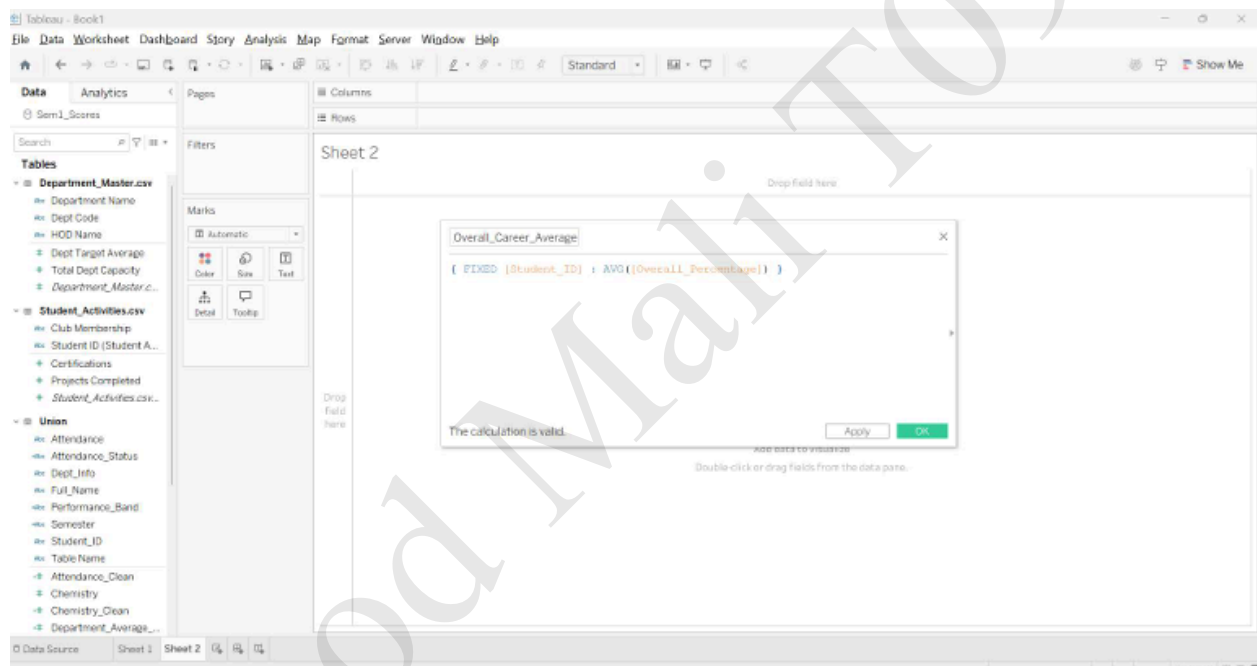
o Field Name: Student_Overall_Career_Average

o Tableau Formula:

Code snippet

{ FIXED [Student_ID] : AVG([Overall_Percentage]) }

o Click OK.



3. Set its Number Format to Percentage (1 decimal place, Just like we did in previous step).

Step 3: Calculate Performance Variance (Student vs. Department Benchmark)

1. Right-click in the Data Pane --> select Create Calculated Field....

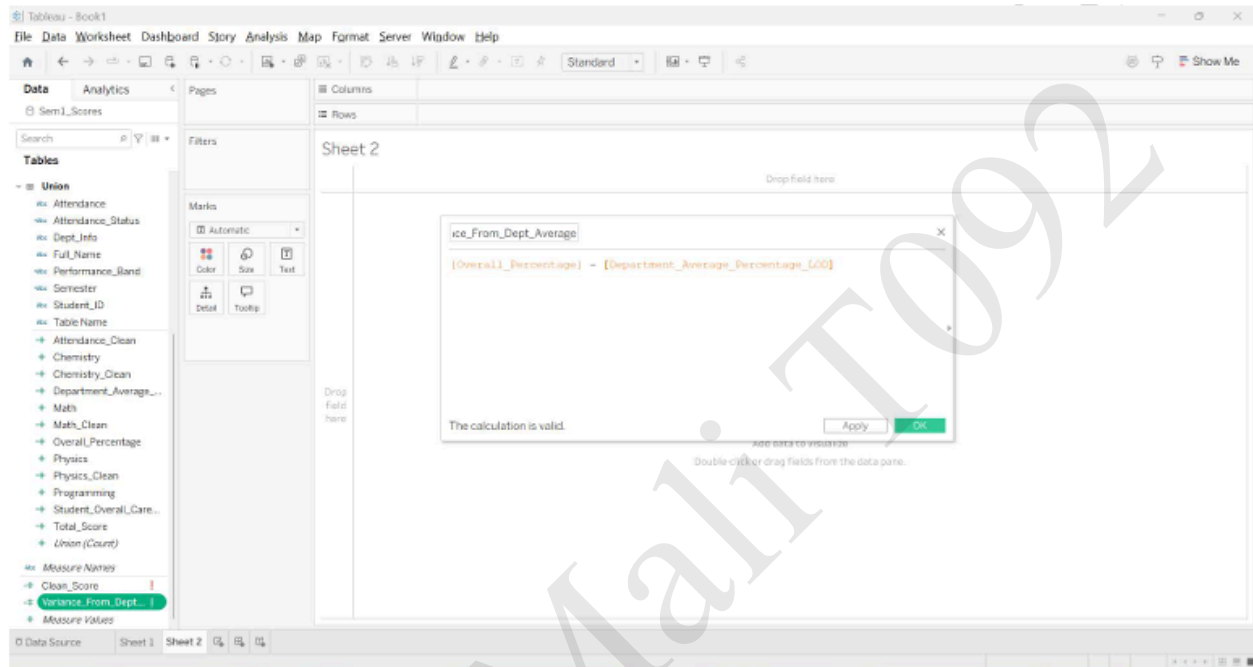
2. Create the relative difference metric:

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- o **Field Name: Variance_From_Dept_Average**
- o **Tableau Formula: Code snippet [Overall_Percentage] - [Dept_Average_Percentage_LOD]**
- o **Click OK.**



3. Set its Number Format to Percentage (1 decimal place).

1 Step 4: Create a Benchmark Indicator Dimension

1. Right-click in the Data Pane -->select Create Calculated Field....

2. Create an automated comparison label:

o **Field Name: Benchmark_Status**

o **Tableau Formula:**

Code snippet

IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark" ELSE "Below Department Benchmark"
END

o **Click OK.**

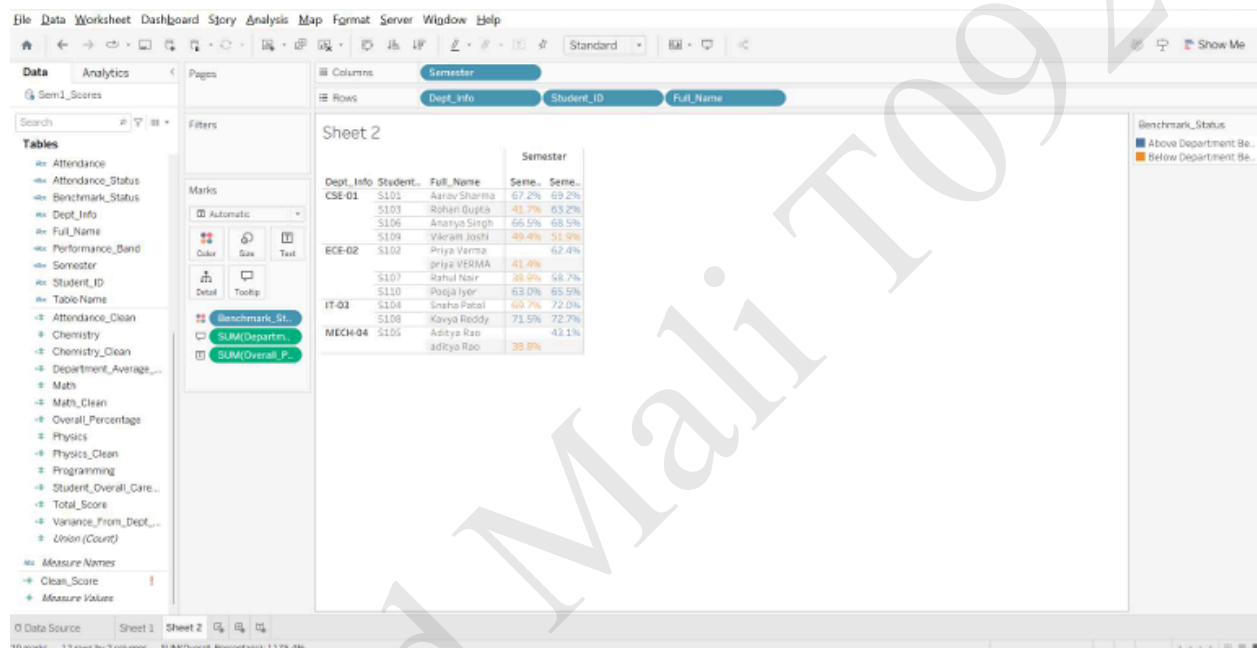
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Step 5: Build the LOD Benchmark Visualization

- 1. Drag Dept_Info, Student_ID, and Full_Name to the Rows shelf.**
- 2. Drag Semester to the Columns shelf.**
- 3. Drag Overall_Percentage onto Text on the Marks card.**
- 4. Drag Dept_Average_Percentage_LOD onto Tooltip on the Marks card.**
- 5. Drag Benchmark_Status onto Color on the Marks card.**



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G: Apply Advanced Data Cleaning Techniques

Step 1: Clean Leading/Trailing Whitespace and Inconsistent Name Casing

- 1. Navigate to Sheet 1 (or a new worksheet).**
- 2. Right-click an empty space in the Data Pane --> select Create Calculated Field....**

- 3. Create the trimmed and standardized student name field:**

- o Field Name: Full_Name_Clean**

- o Tableau Formula:**

Code snippet

```
UPPER(LEFT(TRIM([Full_Name]), 1)) +  
LOWER(MID(TRIM([Full_Name]), 2, FIND(TRIM([Full_Name]), " ") -  
1)) + " " + UPPER(MID(TRIM([Full_Name]),  
FIND(TRIM([Full_Name]), " ") + 1, 1)) +  
LOWER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") +  
2, LEN(TRIM([Full_Name]))))
```

- o (Alternatively, for standard trim-only cleaning):**

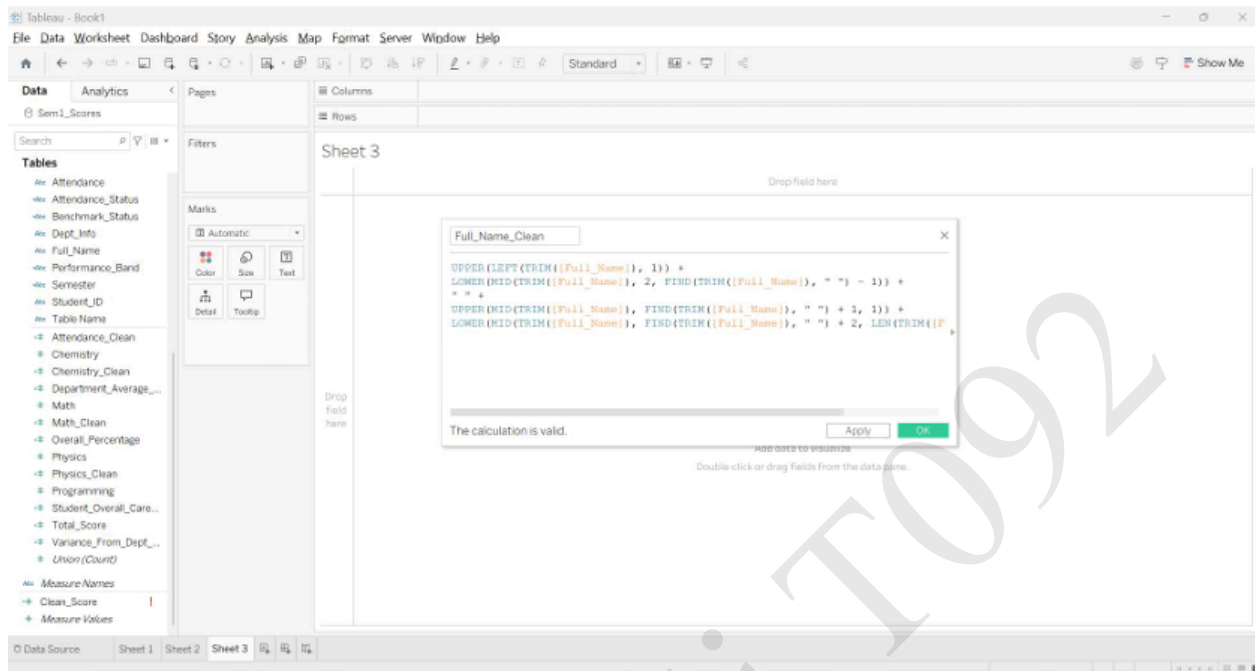
Code snippet TRIM([Full_Name])

- o Click OK.**

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Step 2: Parse First Name and Last Name using SPLIT()

1. Extract the First Name:

o Right-click in the Data Pane --> select Create Calculated Field....

o Field Name: First_Name

o Tableau Formula:

Code snippet

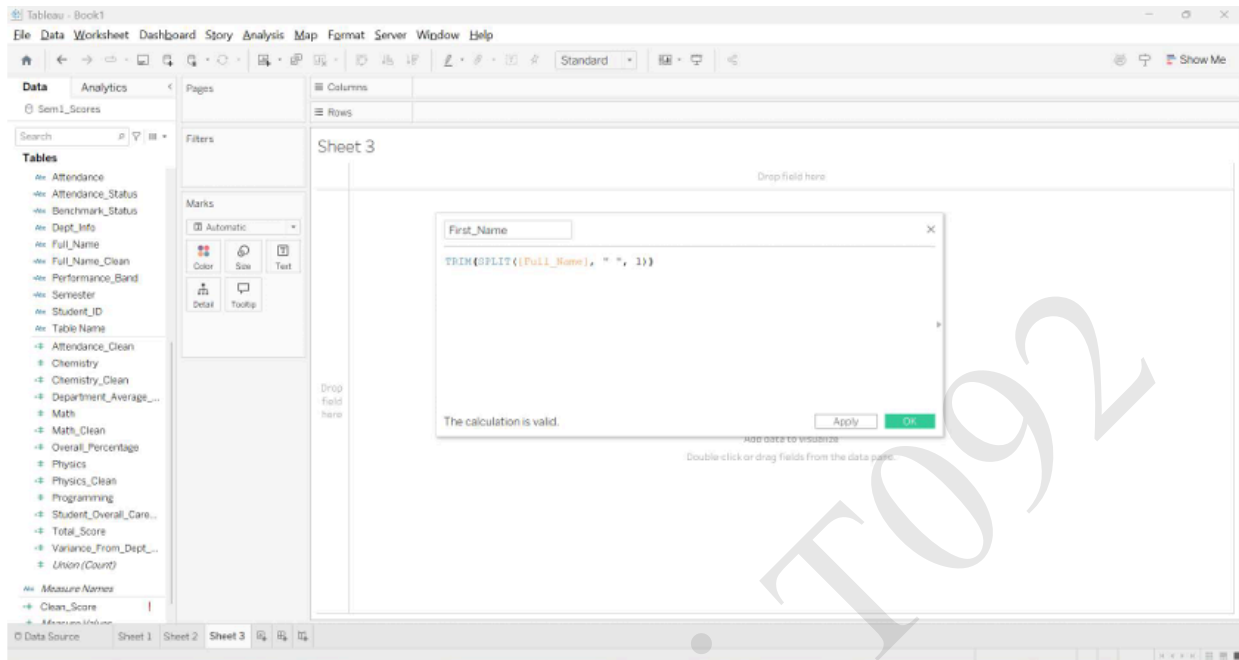
TRIM(SPLIT([Full_Name], " ", 1))

o Click OK.

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Step 3: Split Composite Department Codes (Dept_Info)

The column Dept_Info contains combined strings like CSE-01, ECE-02, and MECH-04. Separate them into distinct dimensional attributes:

1. Extract the Department Acronym:

o Right-click in the Data Pane --> select Create Calculated Field....

o Field Name: Dept_Branch o Tableau Formula:

Code snippet SPLIT([Dept_Info], "-", 1)

o Click OK. 2. Extract the Batch / Section ID:

o Right-click in the Data Pane --> select Create Calculated Field....

o Field Name: Section_ID

o Tableau Formula:

Code snippet

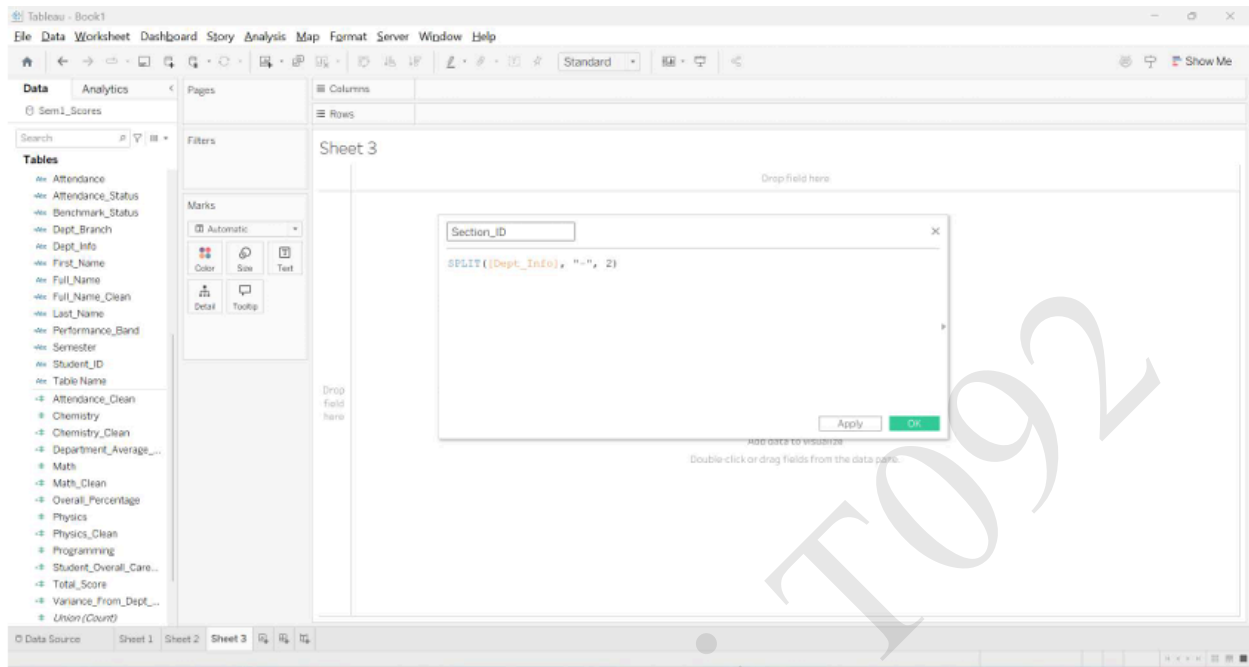
SPLIT([Dept_Info], "-", 2)

o Click OK.

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Step 4: Create a Clean Composite Student Label

Combine cleaned names, student IDs, and department codes for clean dashboard titles and search filters:

- 1. Right-click in the Data Pane --> select Create Calculated Field....**
- 2. Field Name: Student_Display_Label**
- 3. Tableau Formula:**

Code snippet

[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"

- 4. Click OK.**

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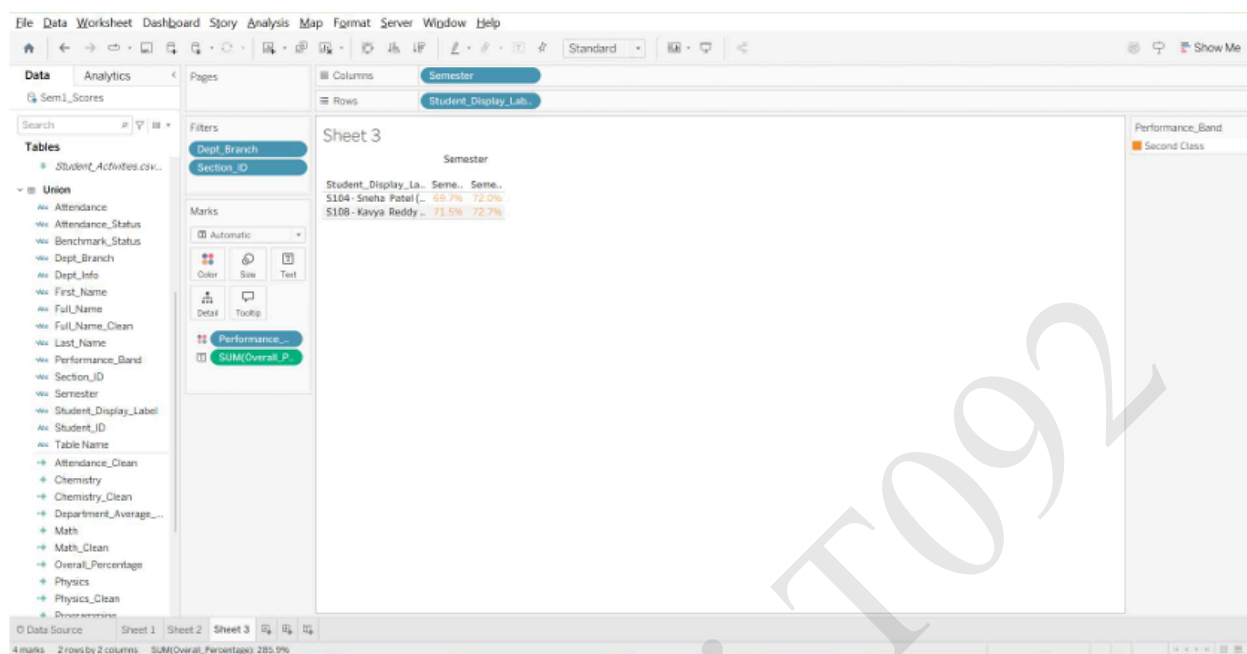
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The screenshot shows the Tableau Desktop interface. The top menu bar includes File, Data, Worksheet, Dashboard, Story, Analysis, Map, Format, Server, Window, and Help. The main workspace is titled 'Sheet 3'. On the left, the 'Data' pane shows a hierarchy of fields: Sem1_Scores, Search, Tables, and Marks. The 'Marks' card is selected, and the 'Automatic' dropdown is open, showing options like Color, Size, Text, Detail, and Tooltip. The 'Columns' and 'Rows' shelves are empty. A dialog box titled 'Student_Display_Label' is open, showing the calculated field formula: `{Student_ID} + " - " + {Full_Name_Clean} + " (" + {Dept_Branch} + ")"`. The dialog also includes an 'Apply' button and a 'OK' button. The status bar at the bottom shows 'Data Source' and 'Sheet 1 | Sheet 2 | Sheet 3'.

1. Place `Student_Display_Label` on the Rows shelf.
2. Place `Semester` on the Columns shelf.
3. Place `Overall_Percentage` on Text.
4. Place `Performance_Band` on Color.
5. Place `Dept_Branch` and `Section_ID` in the Filters shelf. (While Filtering You Can select any of the value you want)

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